

- Root cap can be regenerated

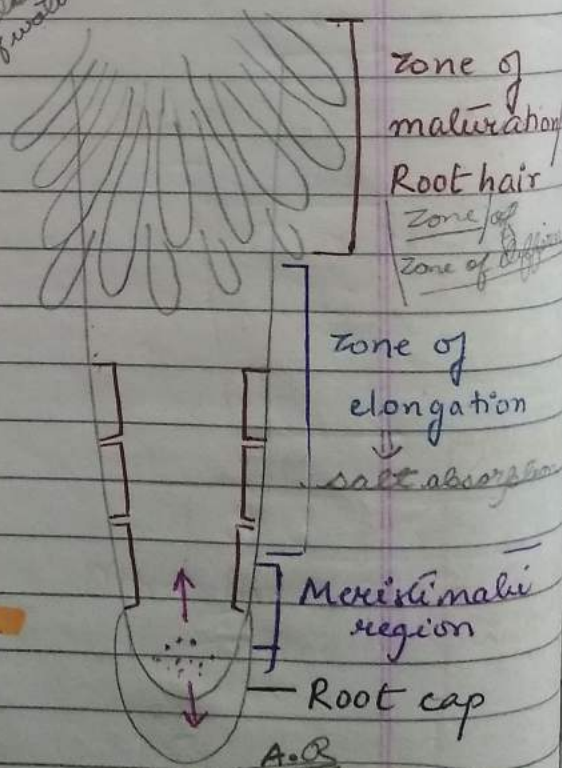
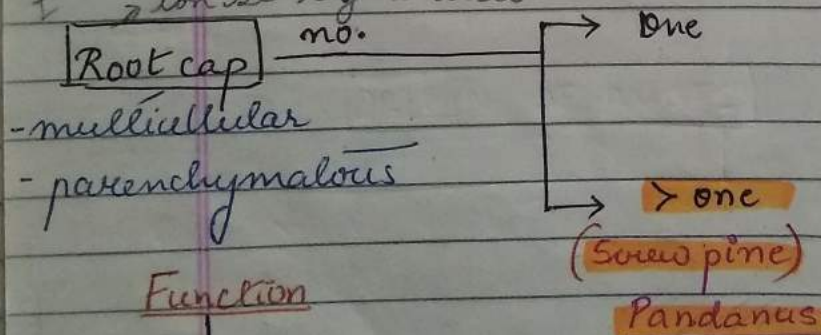
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# MORPHOLOGY OF FLOWERING PLANTS.

## The ROOT

- Non green - underground
- +vely Geotropic
- +vely Hygroscopic. *→ bends towards source of water*
- +vely phototropic

It can be regenerated



- ★ Root cap **not universal**
- Parasitic roots
  - Epiphytic roots.

# Epiphytes

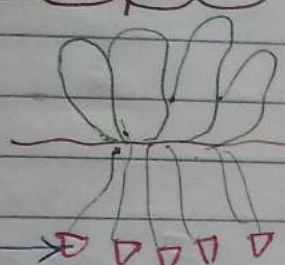
## Epiphytic Roots

- Dependent on the other plant only for attachment.

Space parasites

## Root pocket

e.g. Hydrophytes



→ Balancer - Function.

→ Root pocket regeneration not after injury.

Root pocket

## Zone Of Elongation

### Meristematic

- position - Sub apical
- small, thin cells.
- have dense cytoplasm
- Cell division.
- add cells to both root cap and zone of elongation.

- cell grow w.o.t size
- Large vacuole not

### Functions

- Growth Length.
- Water absp.
- Mineral absp. (Main fun.)

### Maturation Zone

- Differentiation of primary tissue.

- Epidermal (Epidermis)
- Ground tissue
- Cortex
- Pericycle
- Pith
- Vascular tissue.
- xylem
- Phloem

1<sup>o</sup> tissue - young stages of plant.

Differentiation

→ Meristem → Derive

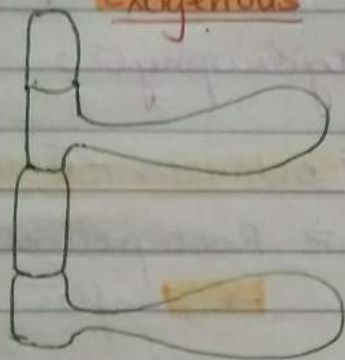
- Cell division capacity lost
- change / Modify special func.

Maturation zone —

Functions

Epidermis  $\Rightarrow$  Root hairs

Exogenous



Pericycle  $\Rightarrow$  Lateral roots

Endogenous

Smaller cells of epidermis

Trichoblast

$\angle$  Capacity to form  
Root hairs

Surface area  $\uparrow$  ces  
Water Absp.

ORIGIN OF ROOTS

Epiderm/  
Cortex.

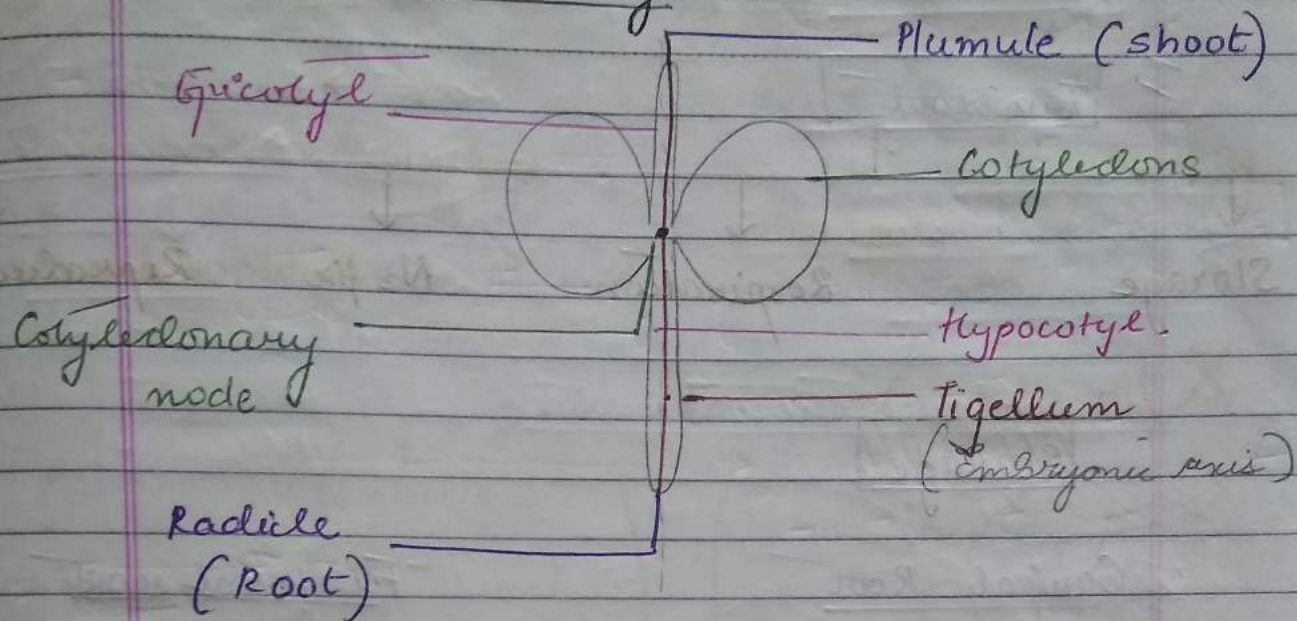
Other than Epidermis/  
Cortex

Exogenous origin

Endogenous origin

- root hairs are present on the rootlets

## Parts OF Embryo



## Types - Root Sys.

Tap root sy.

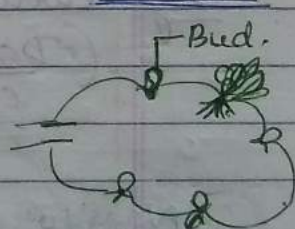
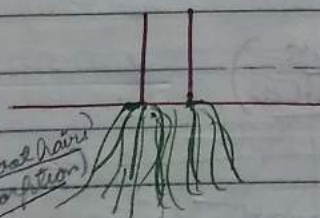
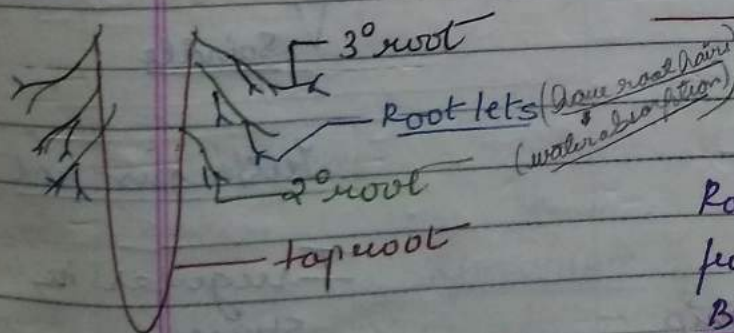
Fibrous root

Adventitious

Origin - Primary root  
↳ elongation of radicle

Primary root  
degenerates

Other than  
the  
Radicle



Roots arise  
from  
Base of stem  
e.g. Monocots

e.g. Bryophyllum

Dicots,  
Monocots, Pteridophytes

Thin, fibrous  
Underground

Thin / Thick -  
Above / Below

e.g. Dicots, Gymnosperms.

Thickness: Thick.

Position: Underground

reduced stem = Acaulescent stem

## Modifications.

### I Tap root

Storage

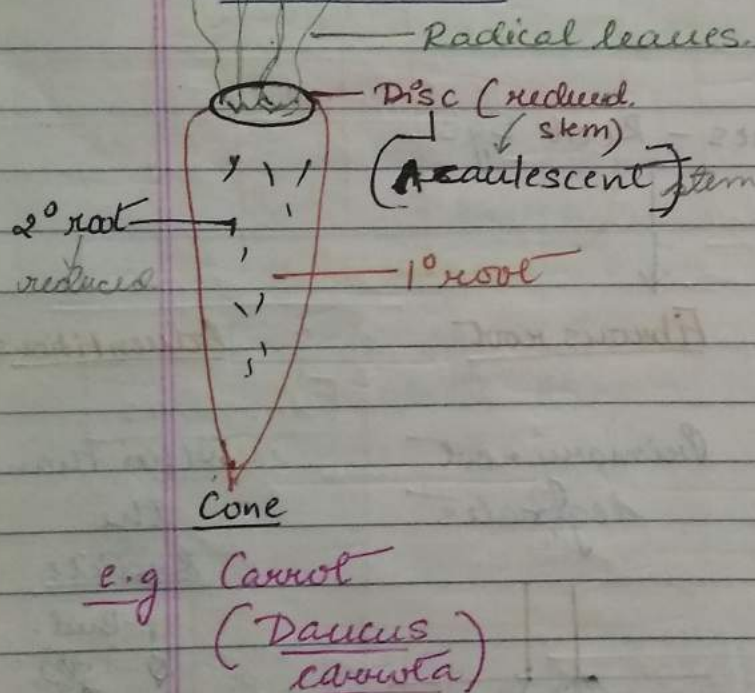
Respiration

N<sub>2</sub> fix

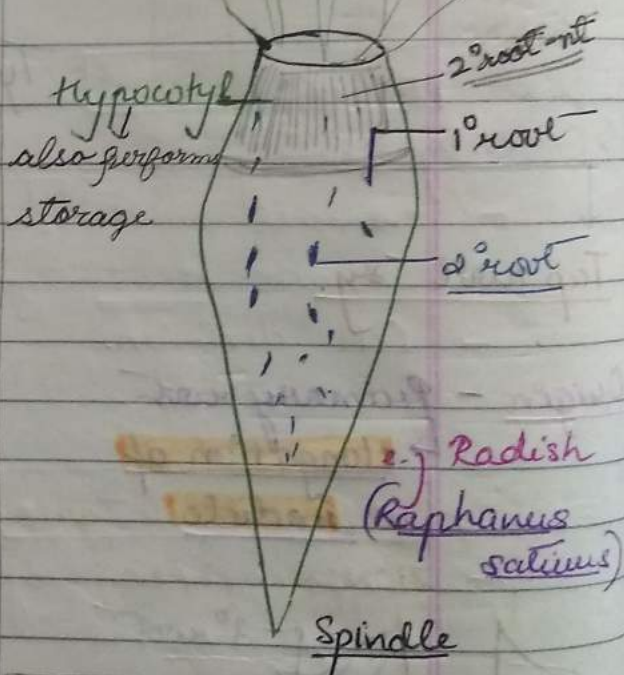
Reproduction

### Storage

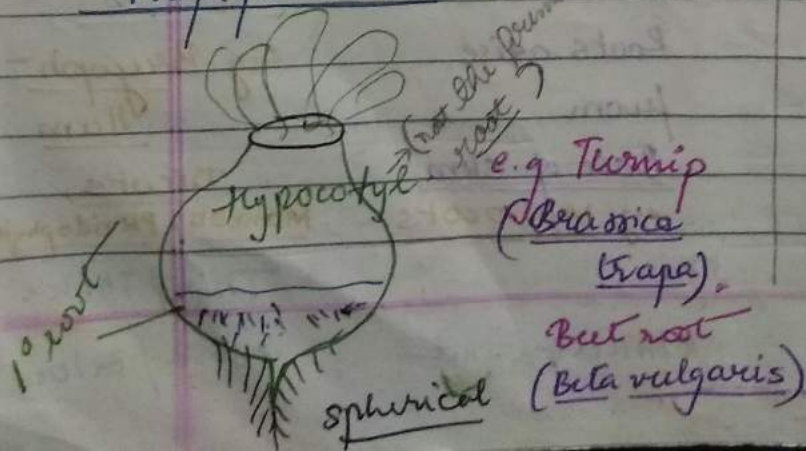
#### Conical Root



#### Fusiform root



#### Napiform root



#### Tuberous root

Irregular in shape.

e.g. Mirabilis jalpa  
(4 o'clock plant)

Respiration — Respiratory roots /  
Pneumatophores /  
Aerophores /

(Eg) Avicennia

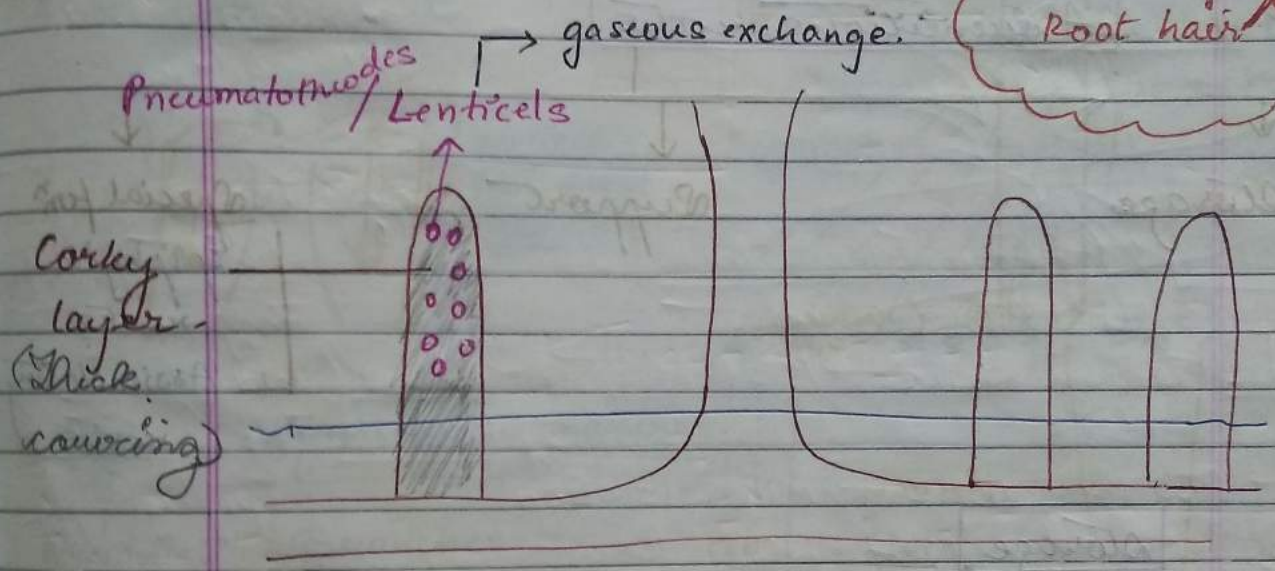
Sonneratia Marshy areas.

Rhizophora  
(Mangroves)

- H<sub>2</sub>O more in soil
- Anaerobic conditions.

Mangrove plants  
— ve geotropic

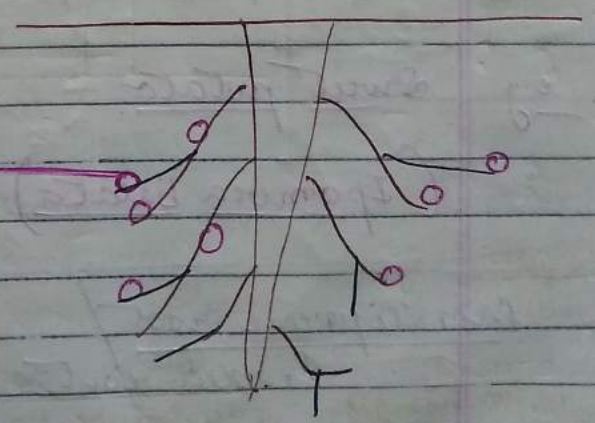
Root cap ~~X~~  
Root hair ~~X~~



N<sub>2</sub> fixation — Nodulated roots (Leguminous roots)

Cicer (Black gram)  
Pisum (Pea)

Root nodules  
↓  
Rhizobium  
↓  
N<sub>2</sub> fix



## Reproduction — Reproductive roots

e.g. Dalbergia

Populus

Root → Bud + nt

→ Radical bud

↓  
New plant

## Auxiliary roots

↓  
Storage

↓  
Support

↓  
Special for  
Epiphytic roots

↓  
Assimilatory roots

## Storage

(i) Tuberous roots

Without definite shape

e.g. Sweet potato

(Ipomoea batata)

(ii) Fasciculated roots

Group of irregular  
tuberous roots

e.g. Asparagus

Dahlia

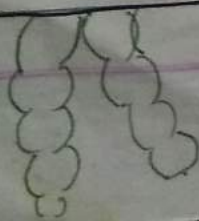
(iii) Moniliform root /  
Beaded root

e.g. Momordica

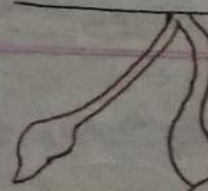
(Bitter melon)

• Portulaca

• Vitis (Grapes)



(iv) Nodulose root



e.g.

• Cucurbita

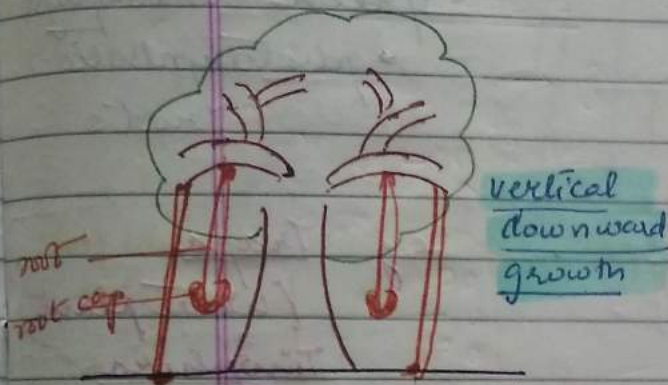
• Turnip

- prop roots, when young are hygroscopic
- Hygroscopic nature → ability to absorb water.

## Support

### Prop roots / Pillar roots

(e.g.) (Banyan tree)  
Ficus benghalensis

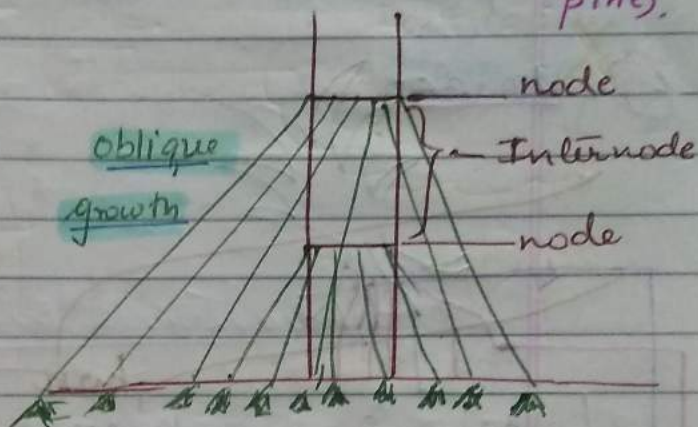


(e.g.) - Hygroscopic nature (Young stages)

Origin - Branches

### Stilt roots / Brace roots

(e.g.) Sugarcane, Maize, Pandanus (Screw pine).



Main stem

### Climbing roots / Climbing

- have weak stems.

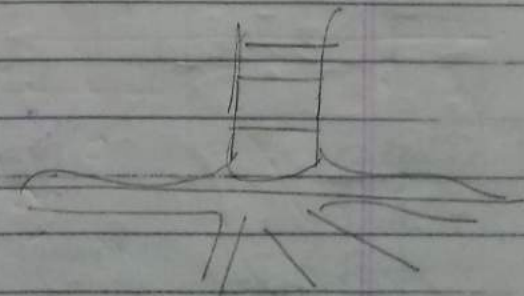
e.g. Money plants,  
(Pothos)

Monstera

Belul (Piper betel)

Black pepper  
(Piper nigrum)

### Buttress roots



Ficus religiosa  
(Peepal)

Terminalia arjuna  
(Arjun tree)

- Epidermis is usually single layered.

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DATE

## Special functions

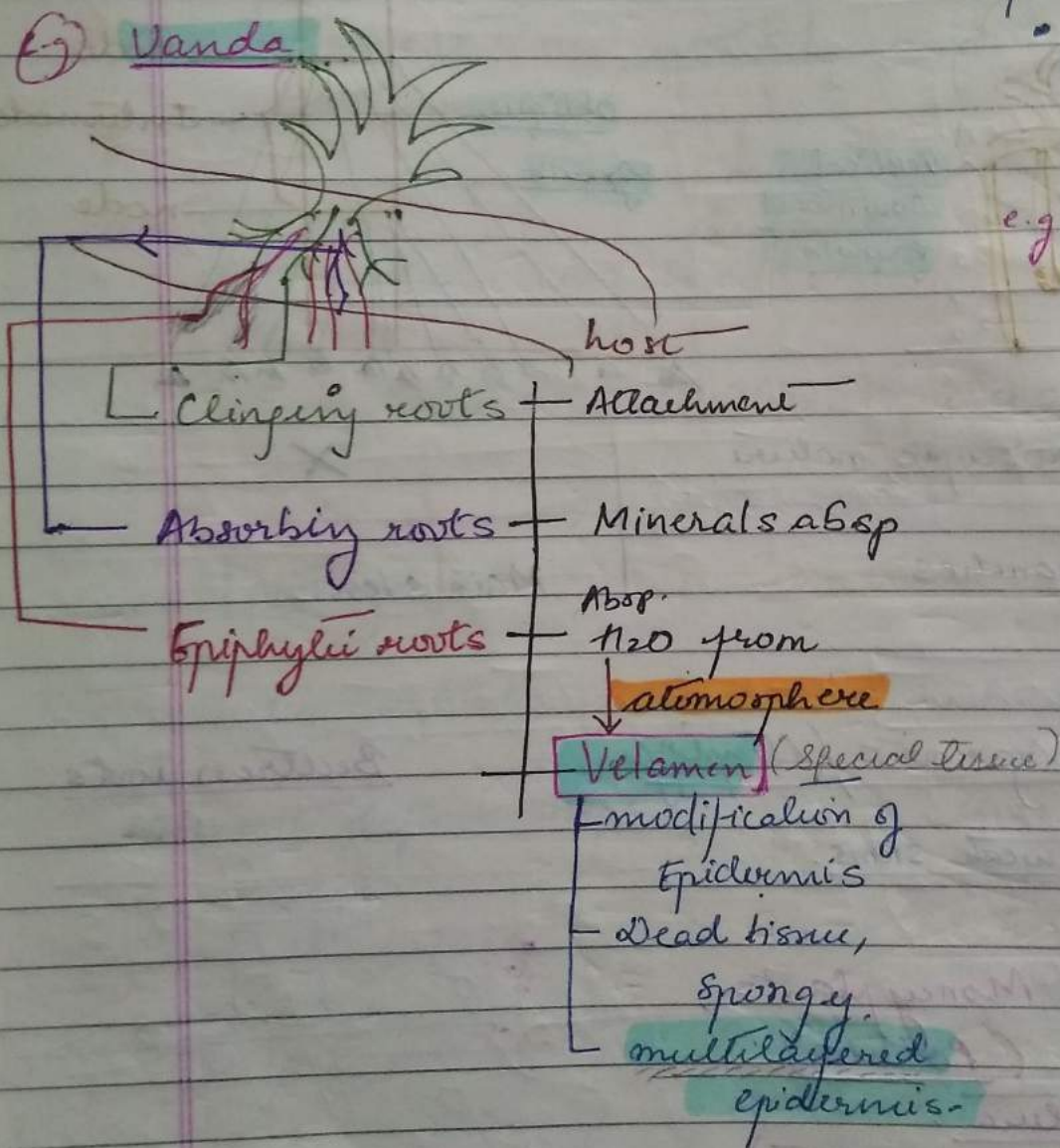
↓  
Epiphytic roots

Space parasites

↓  
Assimilatory roots

- Green - chlorophyll
- photosynthetic roots

e.g. Trapa (water plant)  
Tinospora (woody climber)  
Taenophyllum (Epiphytic plant)

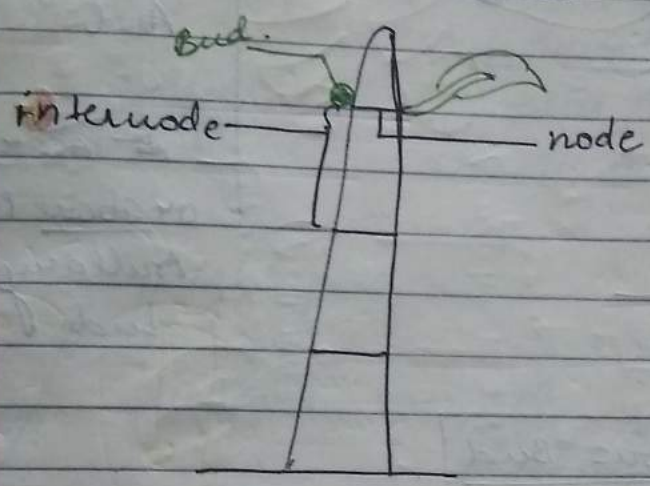


- Haustorial roots → Luscuta, Viscum, Orobanch  
(parasitic roots) - root cap - nt

# The Stem

← Plumule : Part of embryo.

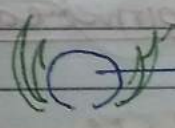
- Above ground → ascending part of plant axis
- Nodes & Internodes.



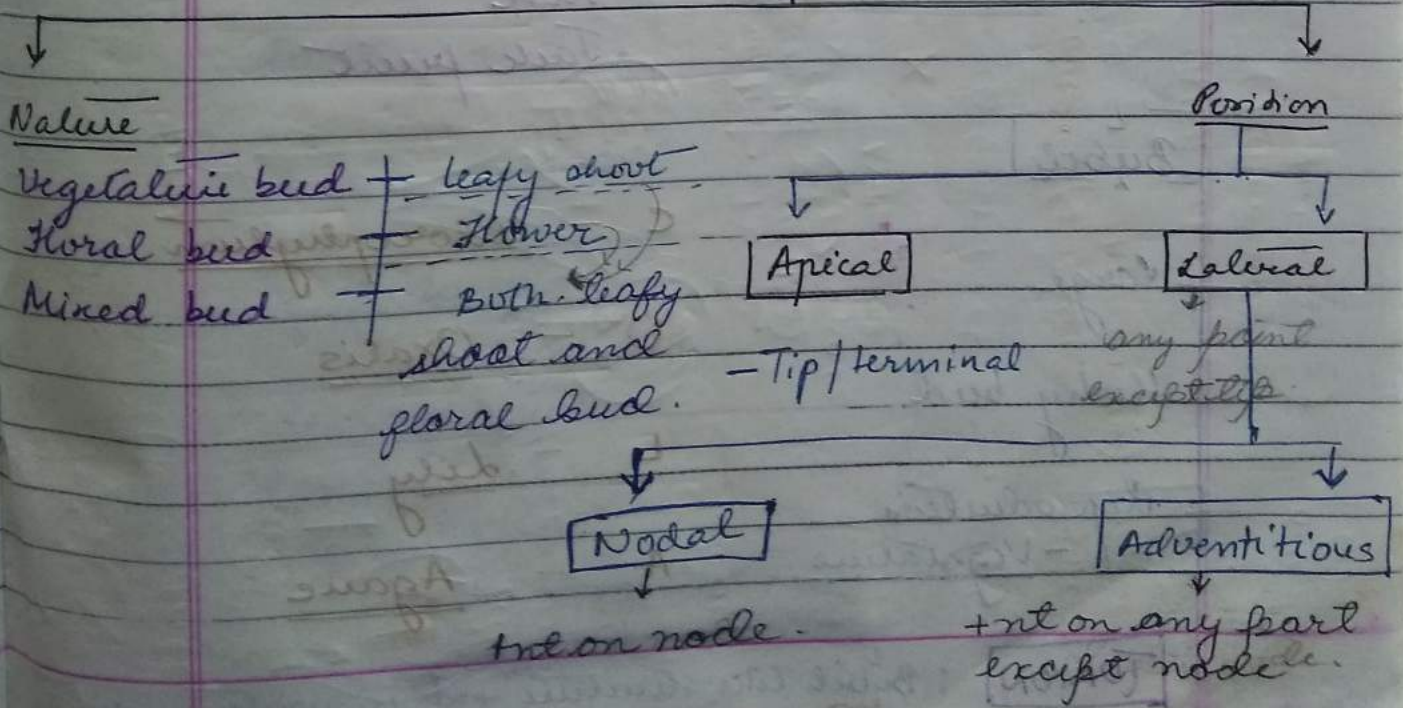
- +vely phototropic
- -vely geotropic
- -ve hygroscopic

## Bud

compact immature shoot  
 immature leaves  
 growing point

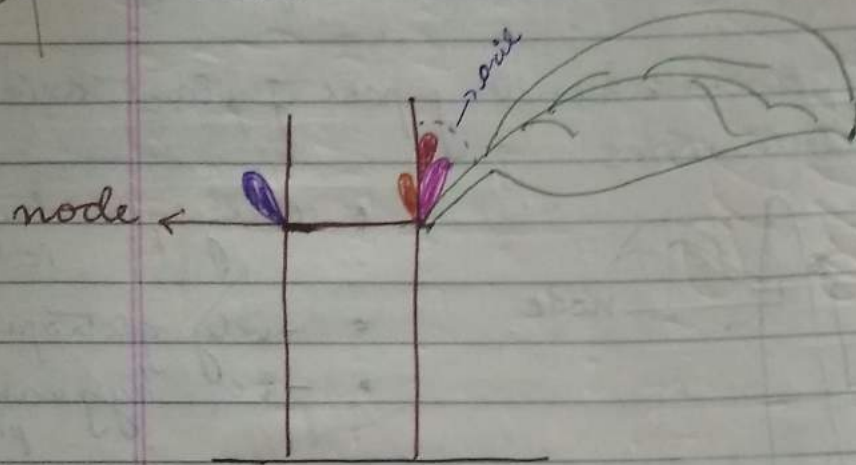


## Types





Bud — Nodal



Axil - Axillary

Away - Extra  
Axil axillary  
bud

A near - Axillary  
above the bud.  
axillary  
bud

- Adventitious Bud

Root

Radical  
Bud

Dahlia, Populus, Dalbergia  
Ipomoea

Leaf

Foliar Bud

Cryptanthum

Stem

Cauline Bud

• Rose  
• Tack fruit

Special Bud

Bubil

Large

fleshy bud

Reproduction

- Vegetative

C - Chlorophyllum

O - Oxalis

L - Lily

A - Agave

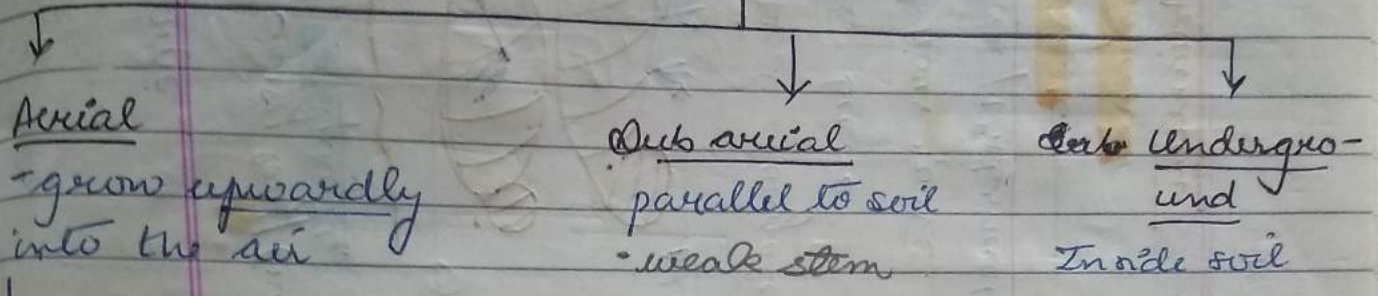
TURION

; Bubil like structure + in water plant

(e.s)

Potamogeton, Utricularia [Blaker Shvo]

## Types of Stem (spread) Ground



- Erect - upright position
- Weak - support required
- Reduced - internodes reduced.



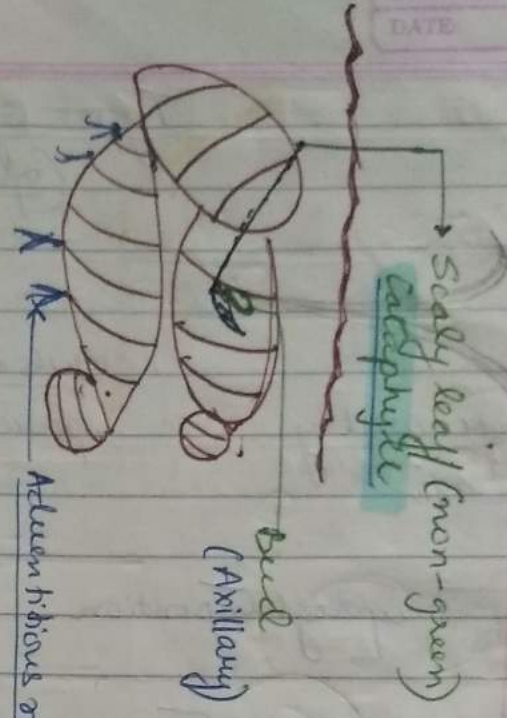
## Modification

- Photosynthesis
- Storage of food
- Reproduction
- Resting/structure
- Perennating

[Bladder Wort]  
Jivorous plant.

• Underground parts - rhizome - underground - rhizome.

# Rhizome



- Condensed elongated
- External buds
- Adventitious roots.
- Branched / Unbranched stem

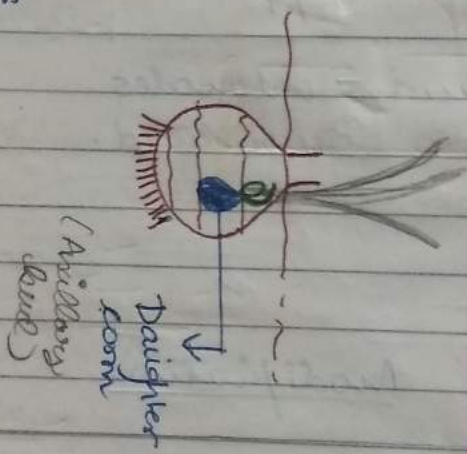
e.g. Ginger  
Banana

(fern). Adiantum (fern)  
↓  
unbranched rhizome

unbranched rhizome

## Underground Modifications of Stem

### Corm

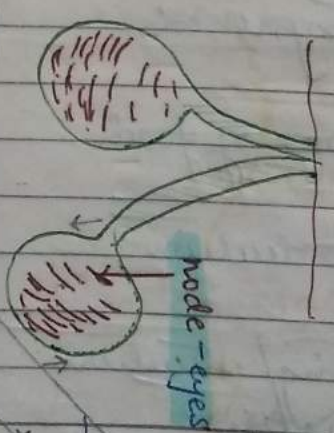


- Condensed spherical
- Ext. buds. ✓
- ✓
- Unbranched

e.g. Coleocasia (variegated)  
Amorphophallus

Fuchsia  
Asclepias  
Trochilium

### Tuber



- Tip - storage
- scaly bud (1-8)
- ✓
- Unbranched

Potato (Solanum)  
Hebe (stem)

Asclepias  
Helianthus (tuberosus)

scapes = above ground



### Bulb

- Reduced, disc
- Internal bud
- ✓
- Unbranched

Onion  
Garlic

Stemmed bulb  
naked bulb.  
Lily.

# Modifications of Stem

leaf  
scape = flower  
pedicel = fruit

## Sub-Aerial

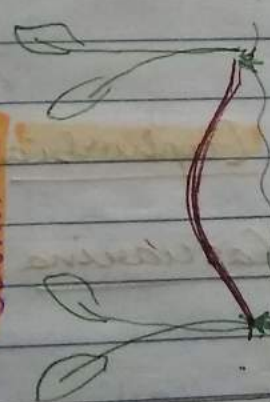
## Modification

Single internode  
runner Offset  
(vegetative)

Pistia, Eichhornia

Stolon

Jasmine



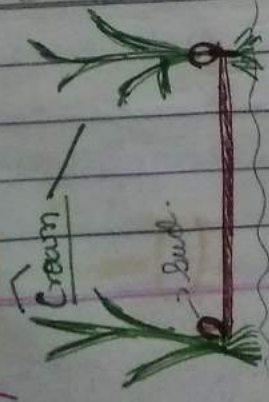
Oblique growth.

> 1

Longest internode

Runner

Grasses



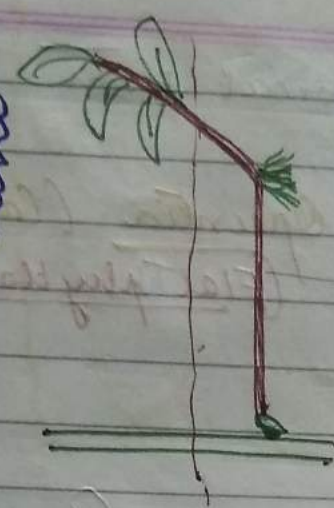
horizontal growth

more than one internode

Sucker

Pineapple, Banana, Chrysanthemum

→ mint



horizontal → Oblique

> 1 internode

stem is present below the ground

Modification - Aerial stem

Phylloclade - e.g. Opuntia (Cactus)  
(Flat phylloclade)

- Xerophytic plants

To save water

↳ leaves reduced - spines

- Function of leaf → Stem  
Green —  
Photosynthesis.

- Whole stem - green

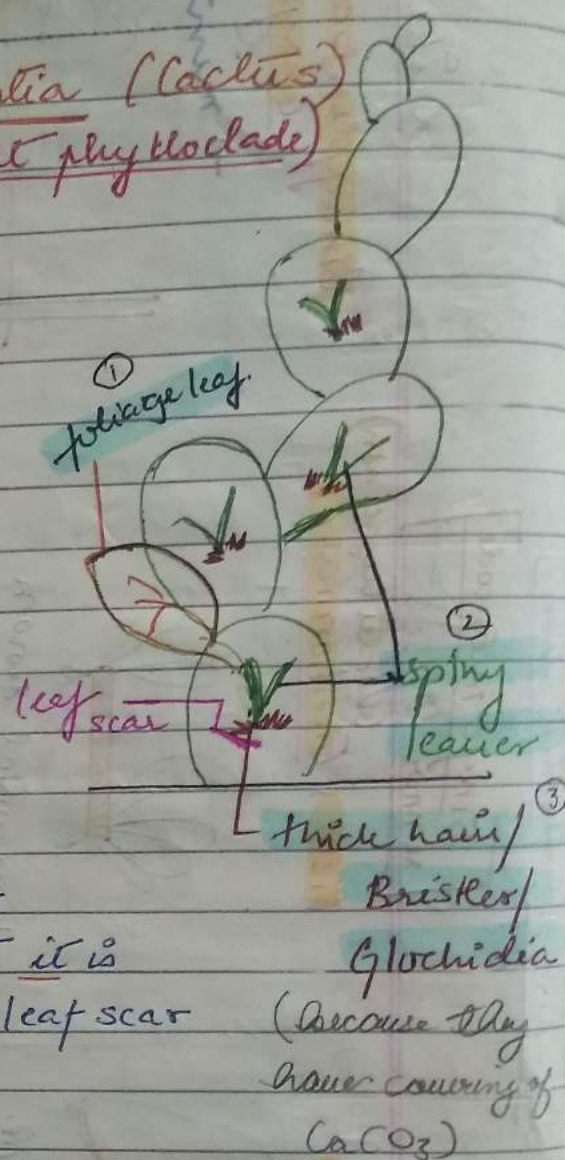
- Unlimited growth.

- Several internodes.

- foliage leaf falls <sup>right</sup> after it is  
formed . & forms leaf scar  
(Caducous leaf)

Cylindrical phylloclade - Euphorbia

- Casuarina



# Comparative

## Cladode

## Phylloclade

## Tendrils

Branches green

whole stem green

• formed from various buds.  
thin, wiry thread like

limited growth

unlimited growth

Coil around the support

one internode

several internodes

Climbing

X

- Foliage

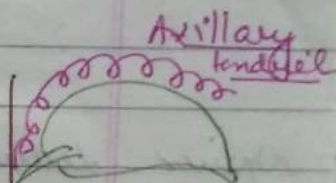
X

- Spiny leaf

X

- Bristles

Extra  
axillary  
tendrils



Asparagus - 1

Ruscus - 2

Internode  
Internode

Apical bud  
tendrils  
Floral  
bud  
tendrils

(e.g)

Pumpkin  
Watermelon  
Cucumber  
Grape Vine

Hard pointed structures



Reduce loss



Protection from  
browsing animals

Thorns, Spines, Prickles, Bristles -

★ In *Bougainvillea*, thorns in the form of hooks are used for climbing.

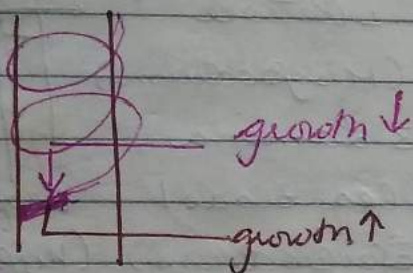
|          | Origin                                              | Vascular Supply of Xylem, Phloem | Corken cells                          | Examples                                                                                            |
|----------|-----------------------------------------------------|----------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------|
| Thorn    | Stem branched                                       | ✓                                | Thick wall used for climbing can scar | <u>Citrus</u> - Hooks<br><u>Bougainvillea</u><br><u>Duranta</u> - leaves<br><u>Alhagi</u> - Flowers |
| Spines   | Leaves                                              | ✓                                | X                                     | <u>Opuntia</u>                                                                                      |
| Prickles | Superficial structure on<br>- Stem<br>- Leaves      | X                                | Easily removed                        | <u>Rose</u><br><u>Smilax</u>                                                                        |
| Bristles | Harder thick<br>- Silica/<br>$CaCO_3$<br>deposition | X                                |                                       | <u>Opuntia</u>                                                                                      |

Potatoes: money plant

## Weak - Aerial stem

### Twining

Sensitive to touch



Differential growth  
coiling

clinging stem X

Convolvulus

### Climbers

Insensitive to touch

No changes in growth

clinging stem. ✓

tendrils -

Cucurbits

Roots - Potatoes

Hooker -

Bougainvillea

★

### Lianas

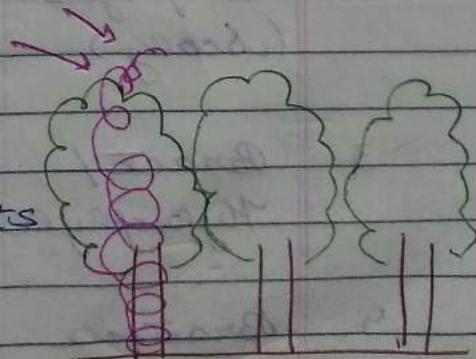
twining, climbers

Woody stems

Perennials

TRF - tropical rain forests

Eg. Hiptage, Tinospora



# The Leaf

- originates from shoot apical meristem (SAM)

- +nt at nodes

- Axil of leaf → Axillary Bud

• Petiole

(Mesopodium)

• Seat of photo-synthesis

3 Lamina

(leaf blade)

Leaf Base

Epipodium (Hypopodium)



## Veins & Veinlets

- Rigidity

- channels for

H<sub>2</sub>O, food

transport

## Types of Leaves

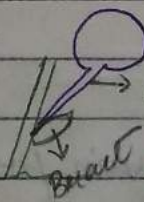
1. Foliage

Green → Fun<sup>n</sup> - Photosynthesis

2. Scalophylls  
(Scaly)

- Dry, non-green → Protection

3. Bract /  
Hypophyll



Pedicel

Involucre (whorl)

4. Bracteole

+nt on the  
stalk of flower i.e. pedicel

Epicalyx (whorl)

5

Sporophylls

Sporangia  
↓

spores

(I) stamen ^ (II) carpel

6

Prophylls

First few leaves formed are different from the rest.

7

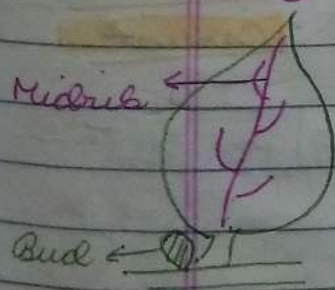
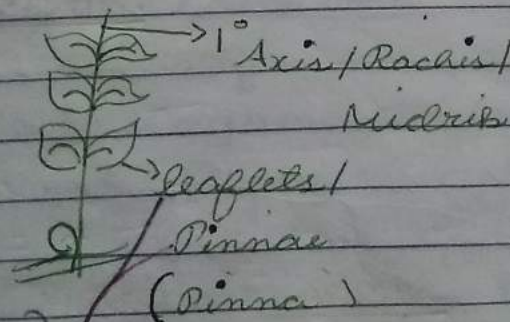
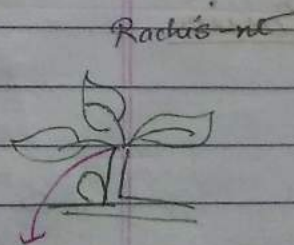
Cotyledonary

Monocot - 1

Dicot - 2

8

All types of leaves present =

PhyllomeTypes of leaves - Lamina  
↓Division of lamina -  
- simplePinnate  
↓Buds x  
Stipule xDivision of lamina -  
- compound  
↓Racemate  
↓

Tip of Petiole

PinnatePalmate

• Leaflets

Born on

Rachis

Type of petiole

Orientation

2 Row

cluster

Architecture

Feather like

Fingers of a palm

Types of Pinnate Leaves

Unipinnate

Bipinnate

Tripinnate

Decomposed

Leaves on

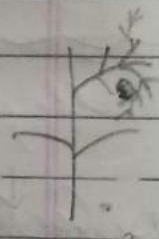
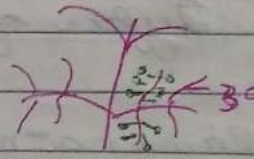
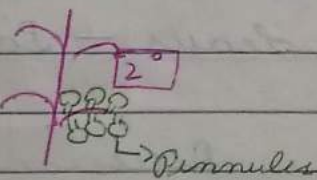
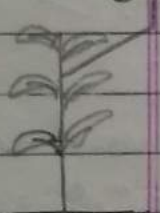
Leaflets on

Leaflets on

Primary axis

2° axis

3° axis

RoseAcaciaMoringa  
(Drum stick)LoranderCarrotNeemMimosaCassia

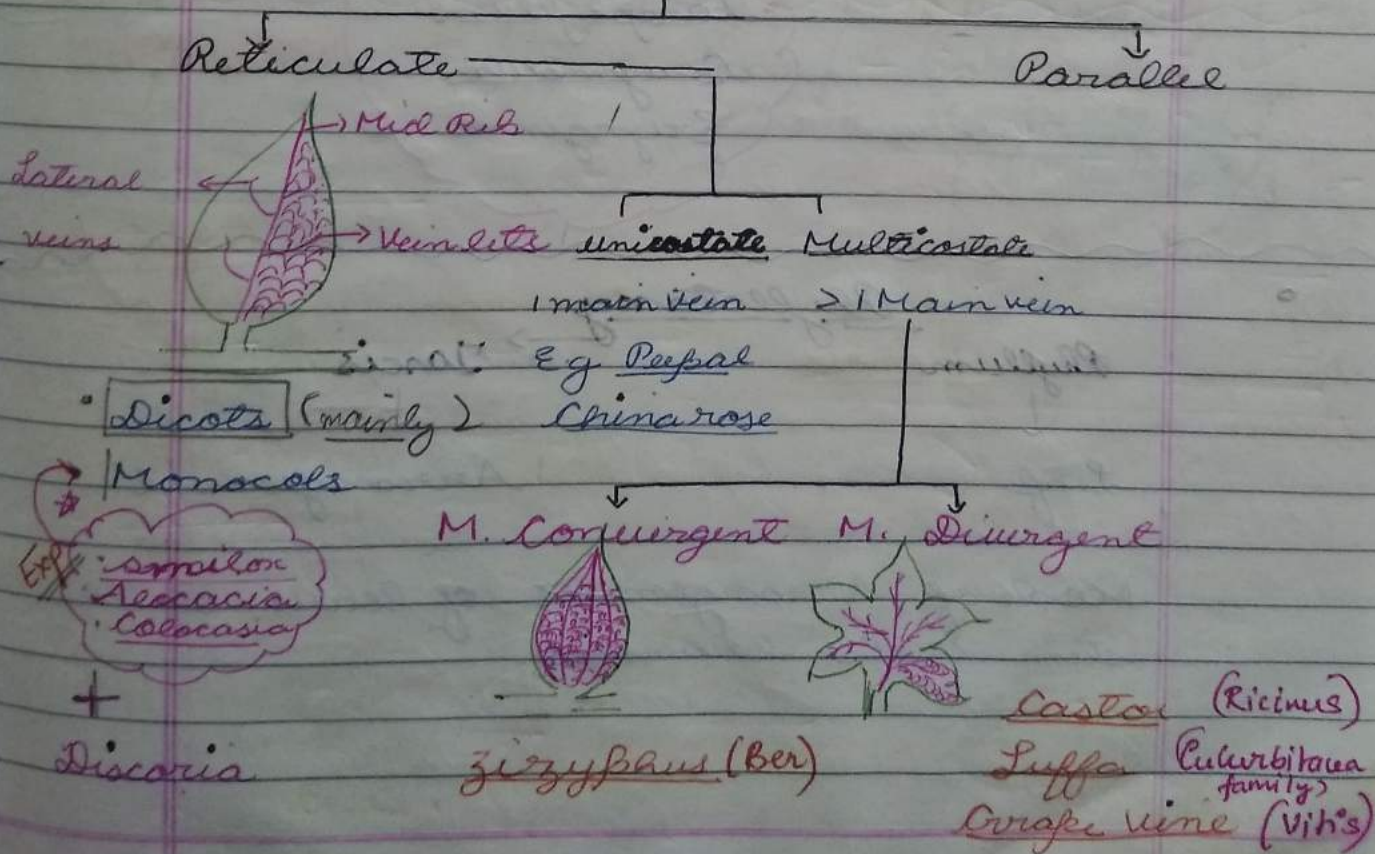
x 2000

x 1000

# Types of Palmate leaves

|             |               |                                                     |
|-------------|---------------|-----------------------------------------------------|
| - 1 leaflet | Unifoliate    | <u>Citrus</u>                                       |
| - 2 leaflet | Bifoliate     | <u>Bignonia</u>                                     |
| - 3 leaflet | Trifoliate    | <u>Trifolium</u> , <u>Dolichosa</u><br><u>Butea</u> |
| - 4 leaflet | Quadrifoliate | <u>Paris</u> , <u>Oxalis</u>                        |
| > 4         | Multifoliate  | <u>Bombax</u> (Red silk cotton tree)                |

## Venation

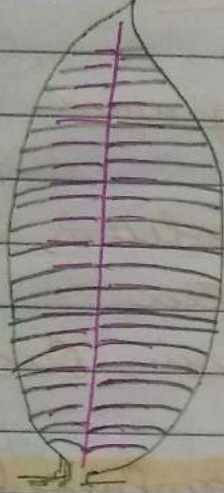


Monocots

Parallel Venation.

- Network - no

↓  
Unicostate



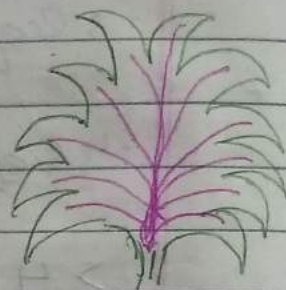
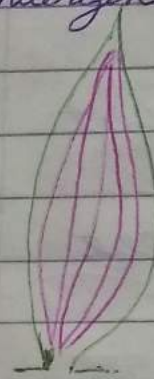
E.g. Banana (Musa)  
Lanna

↓  
Multicostate

↓  
Convergent

↓  
Divergent

- Bamboo
- Grasses



Fan Palm

~~Exp~~

Dicots

Calophyllum  
Corymbium  
Eryngium

Phyllotaxy

Phyllum

↓  
Leaf

Taxis

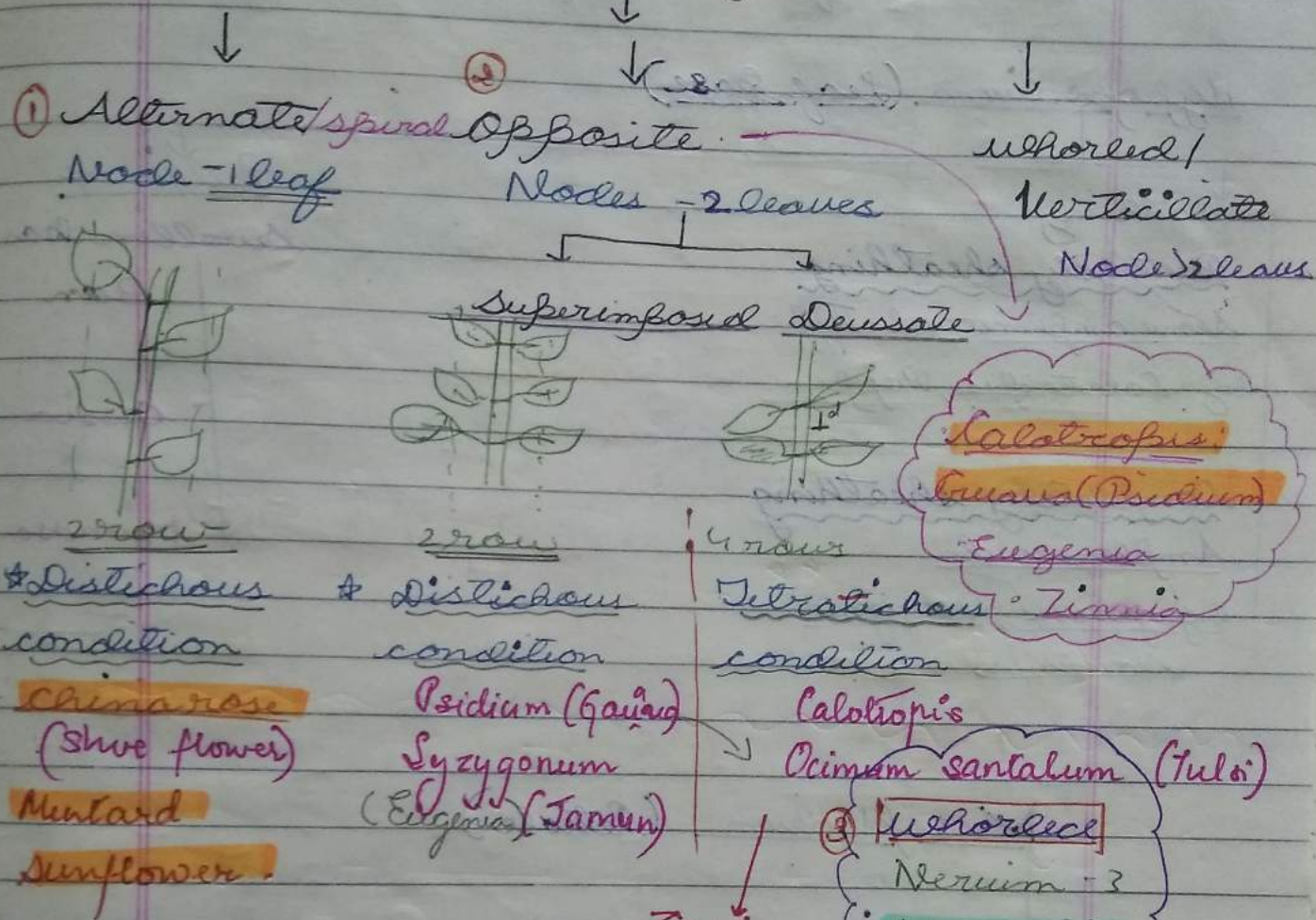
↓  
Arrangement

Mode of Arrangement of leaves on the stem & its branches.

- Alternate type of leaf arrangement also known as spiral phyllotaxy.

DATE

## Phyllotaxy



Zinnia

# Modification of leaves

## Hypophodium (Leaf base)

### Sheathing

Partially sheathing  
Semi amplexicaul type  
e.g. Calotropis, Poppay

Completely sheathing  
Amplexicaul  
e.g. grasses, Polygonum  
Sonchus

### Puttinus

Swollen base

Bean

Cassia

Mimosa

## Mesophodium (Petiole)

Winged (expanded)  
citrus

Spongy (large amount of aerenchyma)  
Eichornia helps in flotation

### Phylloids

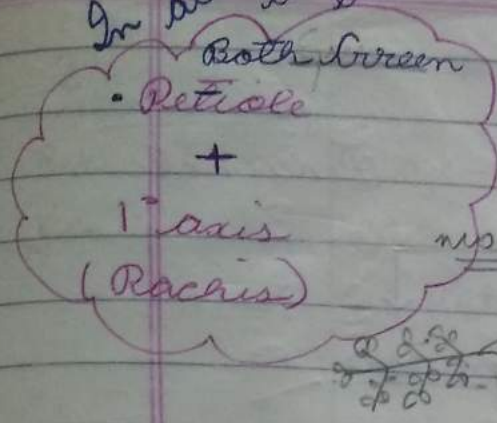
- Xerophytic plants
- Lamina reduction
- Petiole green - Photosynthesis
- Australian Acacia

No of stomata on petiole less as compared to laminae

Hence help prevent water loss.

In acacia has compound leaf

Tendrils perform the function of climbing



\* Phylloclade  
Photosynthetic

Phylloclade  
Photosynthetic

modification Leaf

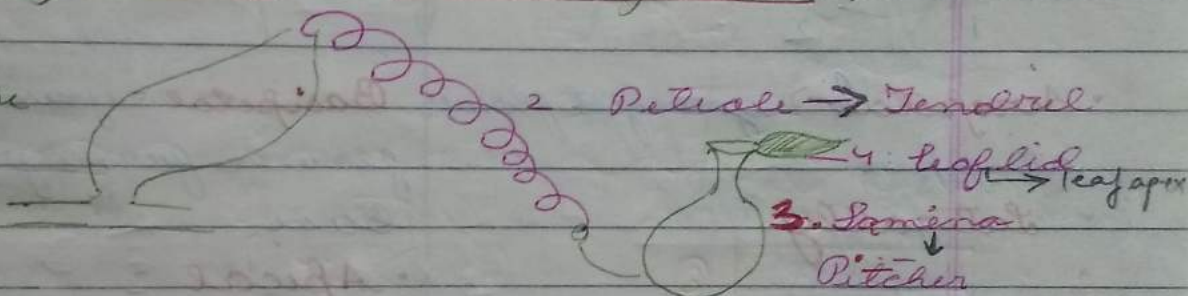
stem

Analogous

## Whole leaf modification

Nepenthes (Pitcher plant)

1. Leaf base green



→ Tendril

Whole leaf : Lalryus sativus

Leaflets : Pisum

Leaf apex : Gloriosa

Petiole

\* Climatis  
\* Nepenthes

modification  
Anacardium  
Analogous  
Legumes

modification  
Leaflets

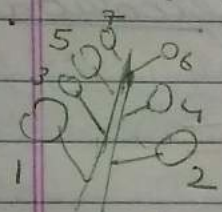
# Inflorescence

Arrangement of flowers →

inflorescence axis  
↓  
Peduncle

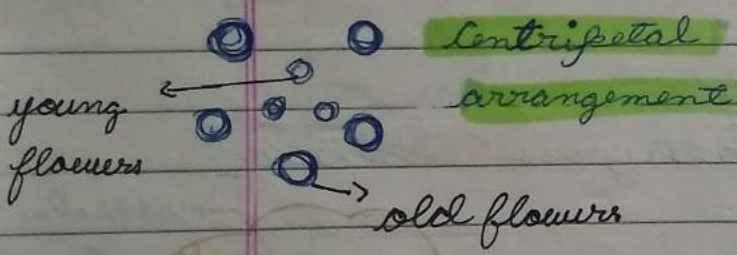
## Racemose

Unlimited growth



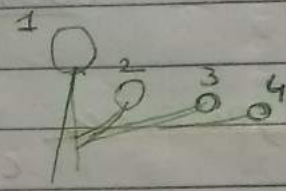
• Acropetal → youngest flower at apex

flowers formed laterally



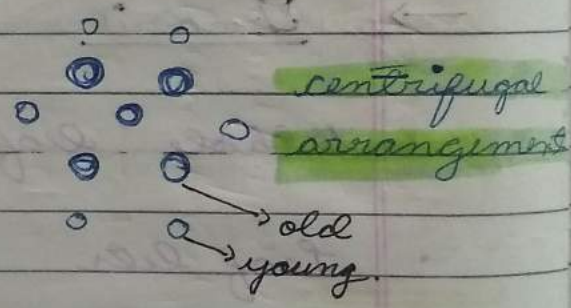
## Cymose

Limited growth



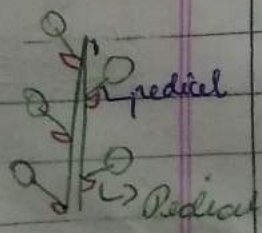
• Basipetal - youngest flower forms from base

• Apical



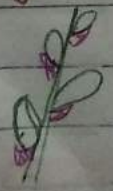
## Types - Racemose

Raceme  
• like spike  
• Racisima



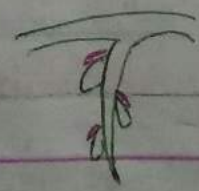
\* Spikes

Achathodia  
Achyranthes



Catkin /  
Amentum

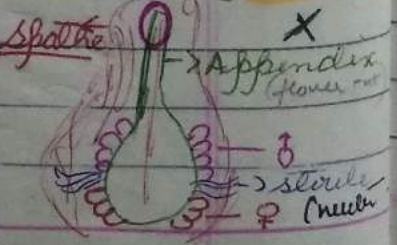
Morus (Mulberry)  
Oak, Birch



(Monocots)

Spadix  
Banana  
Musa, Allocasia

Calocasia, Palm



Acalypha

Spathe

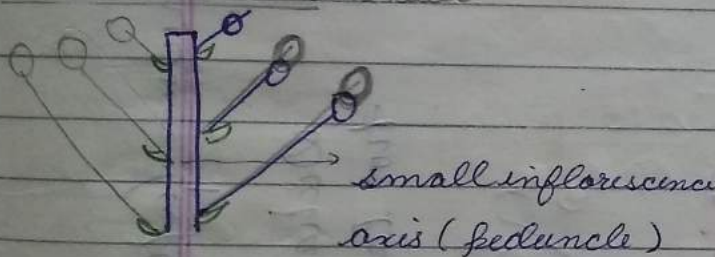
• Capitulum → centrifugal arrangement because it is a type of racemose inflorescence.

DATE: \_\_\_\_\_

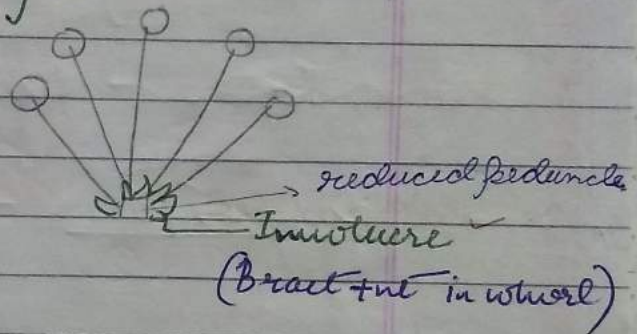
|                                                                  |                                                           |                                                                             |                                                                                                                               |
|------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <p>♀</p> <p><u>Bisexual</u></p> <p><u>Pedicellate flower</u></p> | <p>♀</p> <p><u>Sessile flowers</u><br/>(stalk is -nt)</p> | <p><u>unisexual</u></p> <p><u>sessile</u></p> <p><u>hanging flowers</u></p> | <p><u>Unisexual</u></p> <p><u>sterile Sessile</u></p> <p>• large size<br/>bract surround flowers<br/>called <u>spathe</u></p> |
|------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|

Corymb  
Iberis, Capsella

All the flowers are put in the same level.



Umbel (umbrella like appearance)  
Hydrocotyle



• Compound Corymb  
(branching)  
Cauliflower

Corymbosa xerome - Mustard.

• Compound Umbel  
(branching)  
Coriander

Scapigerous umbel - Onion

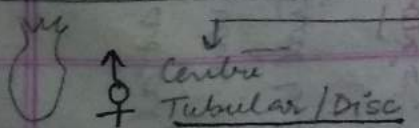
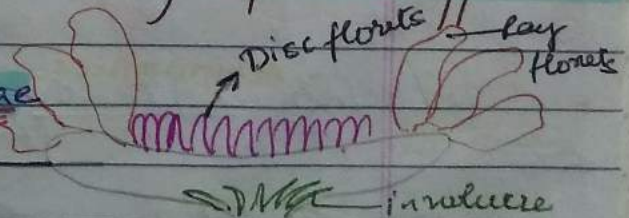
• Capitulum - Head inflorescence - "Single flower appearance"

Zinnia, Cosmos

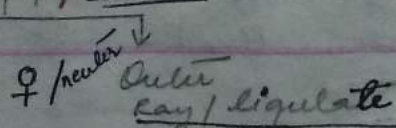
Daisy flower family - Asteraceae

• Disc shaped inflorescence  
axis called receptacle

disk / florets



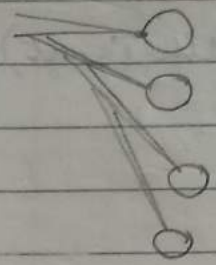
⊕ - Radial Sym.  
- Actinomorphic



- Bilateral Sym.  
- Zygomorphic

o/o

Monochasial / Uniparous.  
One lateral branch

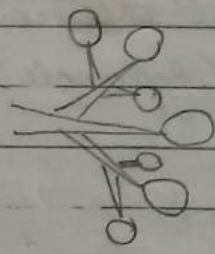


Helicoid  
(Branches arise  
on the same  
side)  
Athyra  
Datura  
Begonia

Scorpioid  
(Alternate  
branch)  
Ranunculus.

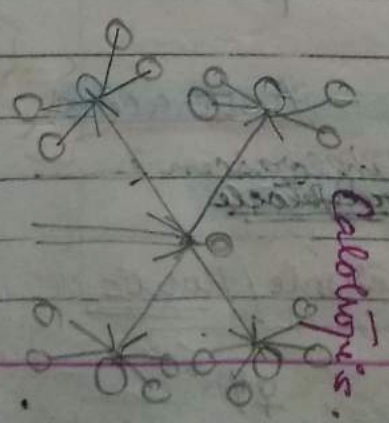
Cymose Inflorescence

dichasial / Biparous  
two lateral branch



Dianthus.

Polychasial /  
Multiparous.  
> 2 lateral branch



Calotropis.

# Special-type Inflorescence

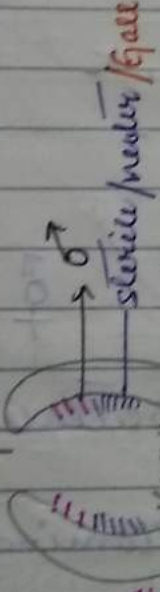
## Hypanthodium

Stipule

Fig - Ficus carica

pepal - Ficus religiosa

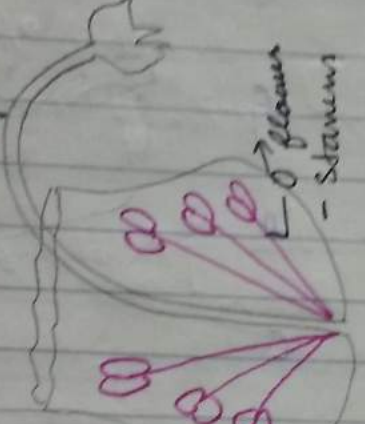
Banyan - Ficus bengalensis



Modified Peduncle

## Cyathium

♀ - Carpel  
flower



Sepals & petals - nt

Bract  
(cup-shaped)

## Euphorbia

• Single flower appearance

Pedunculus

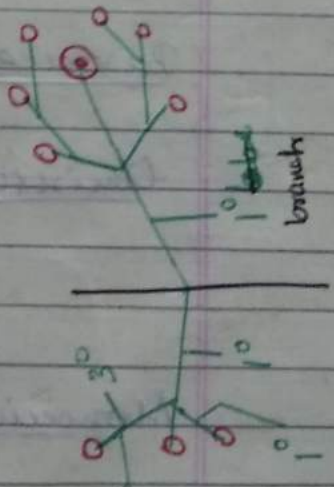
## Achlamydeous

sepals and petals both - nt in male and female flowers

## Verticillaster

Dichasial cyme

with monochasial branches



Ocimum (tulsi)

Salvia (Lamiaceae)

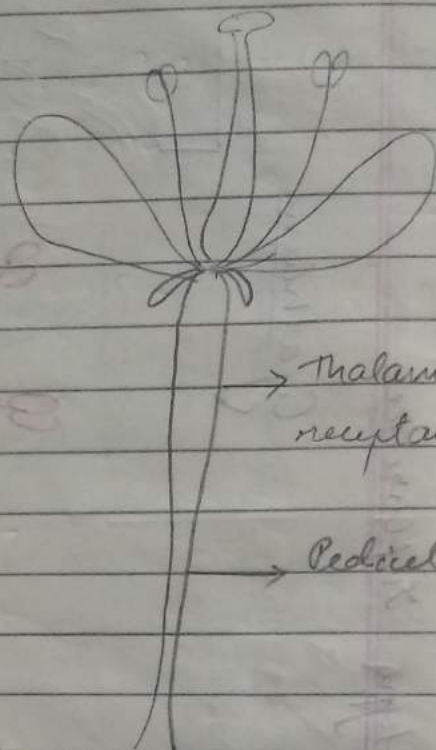
Flower : Defined as highly condensed modified reproductive shoot

- Gynoecium G

- Androecium A

- Corolla C

- Calyx K.



Terms - flower

Complete flower - 4 whorls  
G, A, C, K

Incomplete - missing

Bisexual - G + A, ♀ ♂

Unisexual - G - Pistillate ♀

A - Staminate ♂

Monocious plants - ♀, ♂ same plant

Dioecious

♀, ♂ diff plants

e.g. Coconut  
Cucurbits  
Castor  
Maize  
Alphal

\* Date palm  
\* Mulberry  
\* Papaya.

More common

Polygamous plants

♂, ♀, ♂, neuter flowers

Same plant

e.g: Polygonum

Litchi

Mango.

Dichlamydeous flower

Corolla, Calyx + pet

Monochalamydeous

Perianth

Liliaceae

Achalamydeous flower

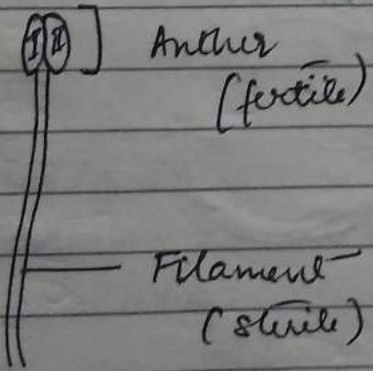
K x C x

Piperaceae family

Calyx -

Androecium

stamens



[2 lobes  
tetrasporangiate]

T = Thalamus G = Gynoecium R = rest of floral parts

## Symmetry - Flower

Actinomorphic ⊕  
Many Vertical planes.

Malvaceae - China rose

Brassicaceae - Mustard

\* Solanaceae - Datura  
chilli

Zygomorphic ⊖  
One vertical plane

Pea } Fabaceae  
Bean }

Guilmoher  
Camia

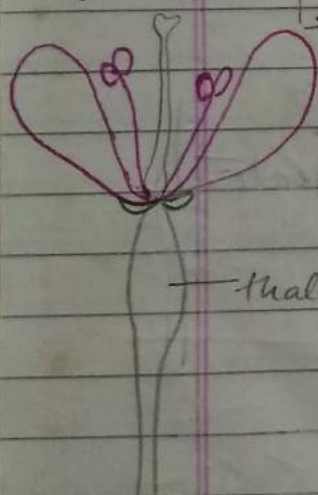
Asymmetrical  
Cannot be divided

Canna

## Position

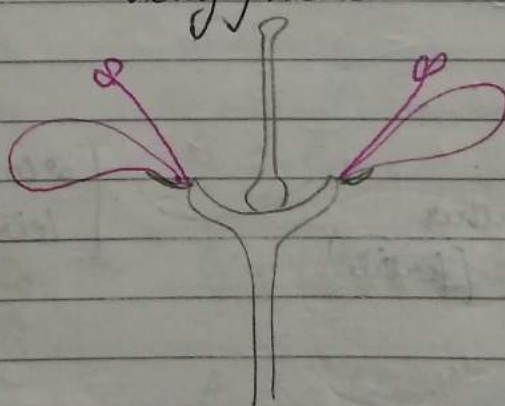
Hypogynous

G

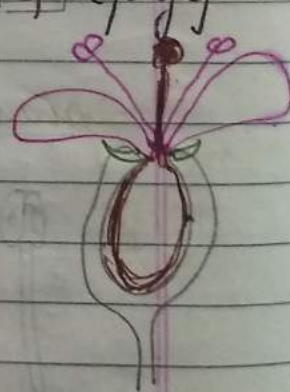


Thalamus

Perigynous.



Epigynous.



\* T - conical

G top / Superior

R - Developing from lower side

China rose

Mustard petunia

T - Expanded

G - Centre Half inferior / Half superior

R - Rim of Thalamus

Peach

Plum

Rose

T - Annular

G - Superior

R - Above it

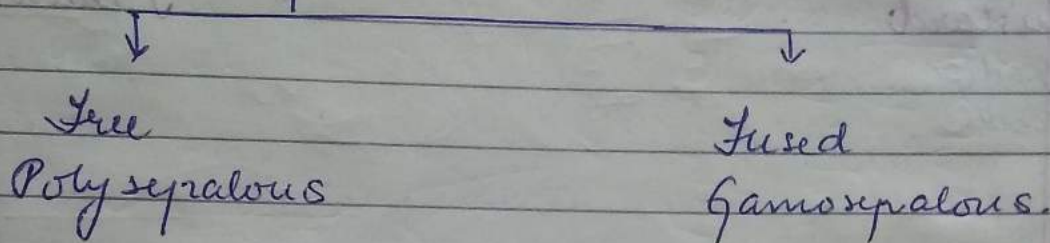
Guava, bilberry

Cucumber

(Sunflower) Ray flowers

## Calyx

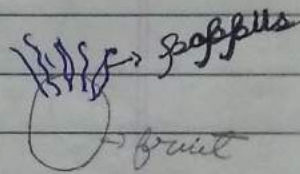
Non-essential whorl  
gamete formation x  
Sepals - Green



### Modification

Helps in  
fruit dispersal

**Pappus** - Sunflower  
Persistent hairy str.



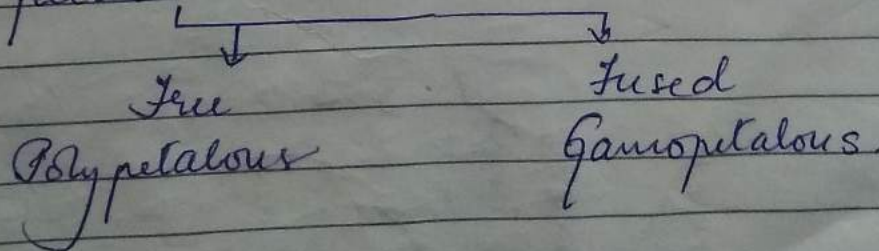
attracts  
insects

**Leafy** - Mussaenda  
One sepal large - white str.

**Spinous** - Trapa  
Persistent - 2 spines  
calyx in the form of

## Corolla

Non-essential whorl.  
petals



# Polypetalous (free)

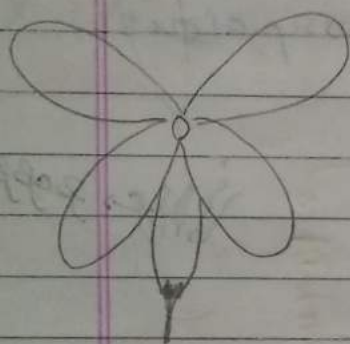


Cruciform  
Brassicaceae

— Mustard

Radish

Turnip



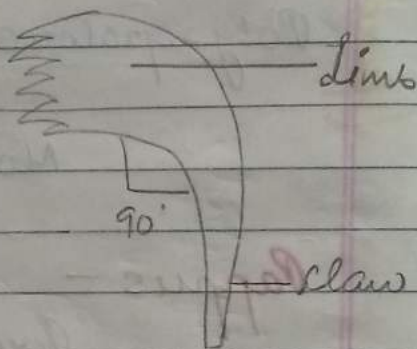
C<sub>4</sub>



Caryophyllaceae

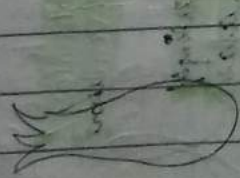
Caryophyllaceae

✓ Dianthus



C<sub>5</sub>

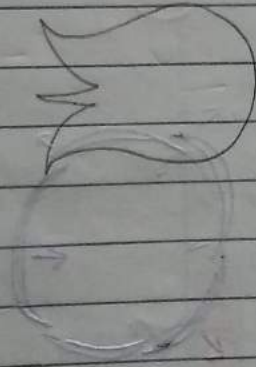
Tubular



Disc flower of  
Sunflower

Companulate

Bell



Physalis

Gamopetalous (fused)

Infundibuliform



Datura

Bilabiate

2 lips  
mouth open

Quercus, Salvia

(Lamiaceae)

Personate

2 lips - very close  
mouth close

Dog flower

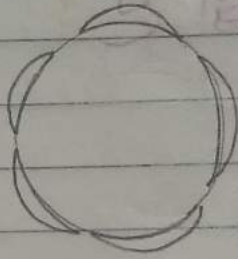
Ligulate

strap

Ray florets

# Aestivation (Arrangement of sepals & petals)

Valvate

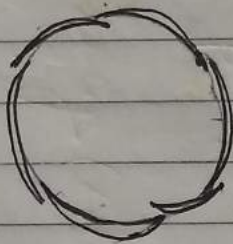


Cucurbitary x

Calotropis

Mustard

Twisted



lady's finger

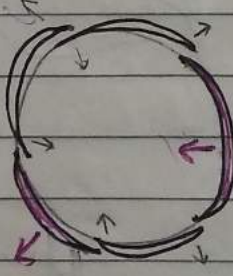
Cotton

China rose

→ Abelmoschus  
esculentus

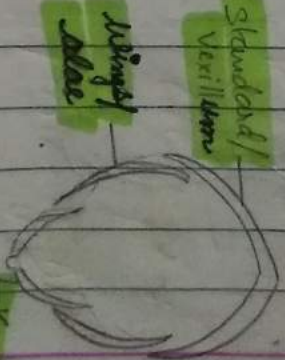
## Aestivation

Imbricate



Cassia

Gulmohur



Standard/  
Vexillum

Wing  
Alae

Pea

Bean

$(1 + 2 + 2)$

(Butterfly corolla)  
Vexillary  
Papilionaceous  
ceus

# Androecium.

## Cohesion - Stamens.

among themselves

### Syngenesious

F - Free  
A - fused  
[A(5)]



Asteraceae

Sunflower

\* Sunflower shows both cohesion and adhesion.

### Adelphous

F - free  
A - free

A - free

Monoadelphous.  
[A(∞)]

Melastomaceae

Chinacrose.

Diadelphous  
[A(9)+1]

Fabaceae

Pea

Polyadelphous.

Leguminosae

(Peas)

Rutaceae

### Synandrous.

F - A - Fusion

Eulubiatis

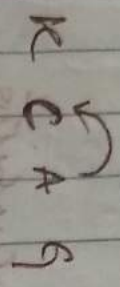
Cucurbitaceae

family

Adhesion - Stamen

Epipetalous

Petals



Chinase rose

Monophloer

Doctanum

↓ Burijal

Epiphyllous / Epithepalous

Persistent (Sepals)



Lily (Lilaceae)

Gynandrous

Gynoecium



Calotropis

fusion of A & G and the

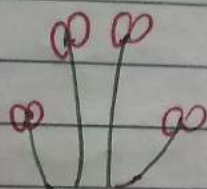
[the Gynoecium] is called.

# Length And Arrangement - Stamen

## Didynamous

4 stamens

- 2 short - outer
- 2 long - inner

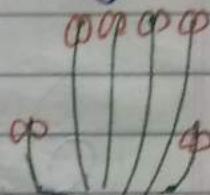


$A_{2+2}$

## Tetradynamous

6 stamens

- 2 - short - outer
- 4 - long - inner



$A_{2+4}$

## Lamiaceae

Ocimum

Salvia

## Brassicaceae

Mustard

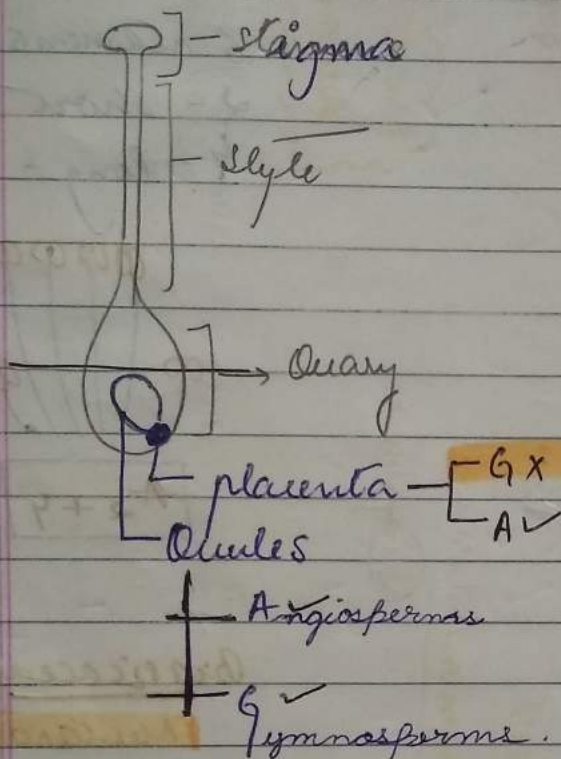
Terms : • Polyandrous  
stamens - free

• Staminode  
sterile - stamen

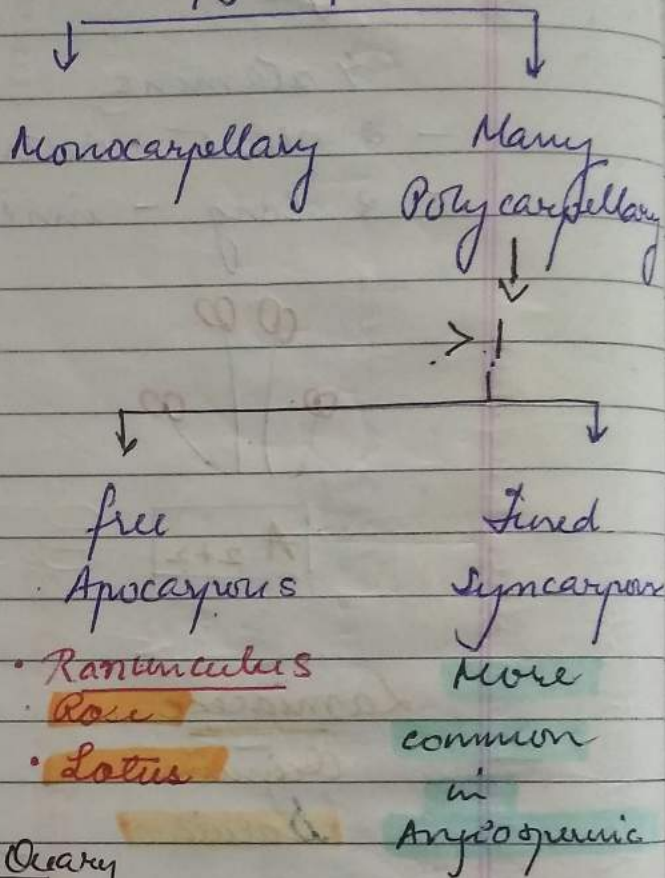
Mustard

# Gynoecium

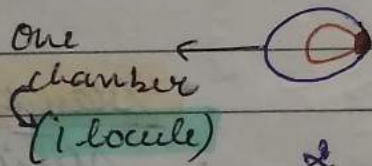
## Carpel



## Gynoecium



## TS of Ovary



1. Unilocular Pea

2. Bilocular Solanum

3. Trilocular Asparagus

4. Tetralocular

5. Pentalocular Ocimum  
Chinarose

> 5

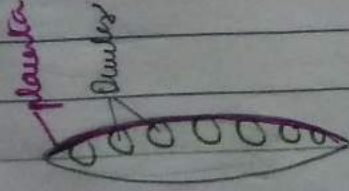
Multilocular

lady's fingers

# Placentation - Arrangement of ovule in ovary

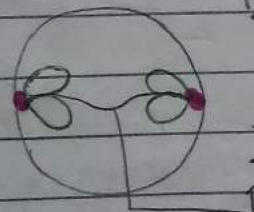
## Marginal

Milobular  
Pea, Bean



## Parietal

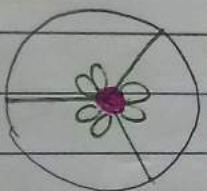
Uni → Bi  
Mustard  
Argemone



false septum  
 (Replum)

## Axile

Bi to Multi  
Lemon  
Lily  
Tomato  
Chinarose

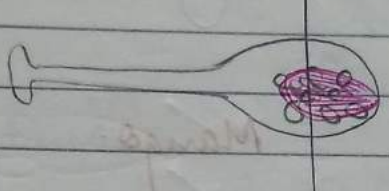


Most  
 Common

seen in  
 angiosperms

## Free Central

Uni  
Dianthus  
Primrose  
 (Primula)  
Stellaria



## Basal

Uni  
MORNA  
Dumbrose  
Wheat  
Boatrace

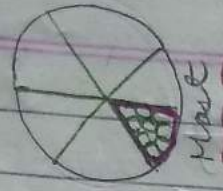


Most  
 Advance

ex  
 - Poaceae  
 - Asteraceae  
 • Single ovule

## Superficial

Multi  
 waterily

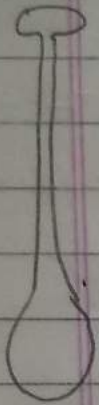


Most  
 Primitive

Large no.  
 of ovules

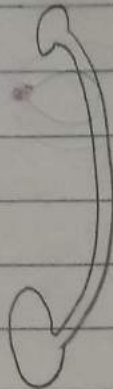
# Position Of style

Terminal



China rose  
(Most common  
in angiosperms)

Lateral style



Mango.

Gynobasic  
from base of ovary



Ocimum  
(Tulsi)

# Symbols - floral formula

1. Br - Bracteate flower

$\overset{\curvearrowright}{C} \ A$  Epipetalous

2. EBr - EBracteate flower (bract - not)

$\overset{\curvearrowright}{G} \ A$  Gynandrous

3.  $\oplus$  - Actinomorphic

$\overset{\curvearrowright}{P} \ A$  Epiphyllous/  
Epitpalous.

4.  $\%$  - Zygomorphic

$G$  - Gynoecium

5.  $\text{♂} \text{♀}$  Bisexual

$G(n)$  = Syncarpous.

6.  $\text{♀}$  Unisexual female

$G_n$  = Apocarpous.

7.  $\text{♂}$  Unisexual male

$\underline{G}_n$  - Superior Ovary

8. K - Calyx

$\overline{G}_n$  - Inferior Ovary

9.  $K_n$  - polysepalous (free)

Epi - Epicalyx (below sepals)

10.  $K(n)$  - Gamosepalous (fused)

● - Posterior side of flower

C - Corolla

Mother axis

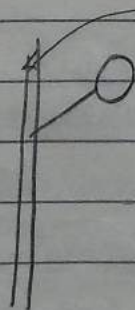
$C_n$  - polypetalous

$C(n)$  - Gamopetalous

P - Perianth

A - Androecium

$A_\infty$  - infinite stamens.



Fabaceae

Solanaceae

Liliaceae

Pea family

Potato family

Lily family

See family

Herbaceous

Habitat

Dicot

Dicot

Monocot

H, S, T

H, S, T

Perennial

Herbs, Shrub, Trees

Herbs

Herbaceous/Woody

Solid / Hollow

✓

Underground

Stem

Tuber (Underground)

stem

(Potato)

Bulb - Onion

Corn - Colchicum

Rhizome - Polygonatum

Reticulate venation

✓

Parallel

Leaf

Simple / Compound

✓

Simple

Phyllotaxy - Alternate

✓

Alternate / Basal

• Pulvius (swollen leaf base)

x

x

• Stipules

x

x

Racemose

Solitary / Cymose

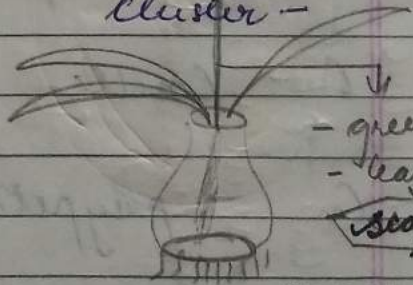
Solitary / cymose

- Umbellate cluster -

Petiole - Petiole

Stipule - Stipule

Triflorous



♂ ♀

♂ ♀

♂ ♀

Scapiflorous

Cymose

Stipule

Flower

# Fabaceae

# Solanaceae

SCNA-PCOFA  
PAGE NO.:  
DATE:

# Liliaceae

Calyx

K(5)

K(5)

P 3+3 / (3+3)

↓  
tube.

Corolla

Vexillary

C<sub>1+2+(2)</sub>

Valvate

C(5)

Androecium

Diadelphous.

A(9)+1

Epipetalous

C(5) A<sub>5</sub>

Epitopalous

P<sub>3+3</sub> A<sub>3+3</sub>

Gynoecium

3G

Monocarpellary  
Unilocular

3G

Bi, Syn.  
Bilvular  
Obliquely

3G

Tri Syn.  
Trilocular

Locusta

Marginal

Axile, Swollen

Axile

Fruit

Legume / Pod

Berry / Capsule

Capsule / Berry  
(rare)

Pulse - Arhar, sem,  
Gram, Soyabean

Food - tomato, potato  
Brinjal

Vegetable -  
Asparagus  
Onion (Allium cepa)

Medicine

Glycyrrhiza  
(Moolali)

Spices - Chillies

Medicine -  
Aloe vera

Oil

- Groundnut,  
Soyabean,

Medicine - Atropa  
belladonna  
Asparagus - Withania  
Somnifera

Ornamental

Tulip,  
Gloriosa

Ornamental - Lupin, Sweet pea

Fumigatory -  
Tobacco

Chemical -  
Colchicum,

Fibres - Sun hemp

Ornamental  
- petunia

Antitumoural  
↓  
colchicine

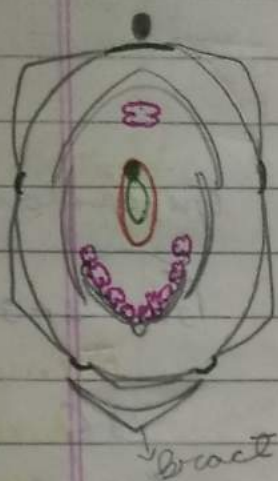
Fodder

Sarbania  
trifolium

- N<sub>2</sub> fixing

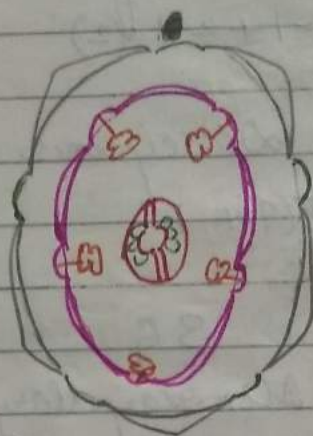
# Labaceae

Br % ♂ K(5) C<sub>1+2+(2)</sub> A(9)+1 G<sub>1</sub>



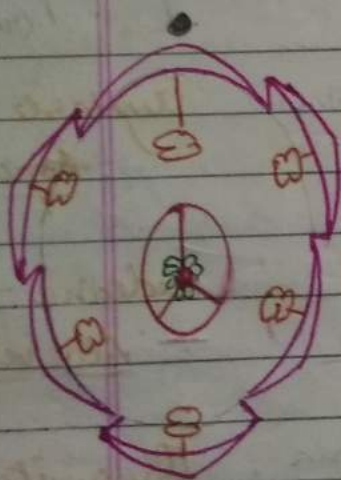
# Solanaceae

♂ % K(5) C(5) A<sub>5</sub>  
G(2)



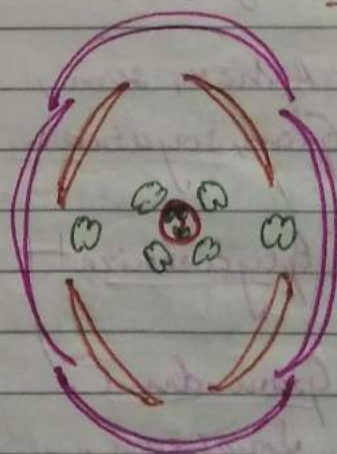
# Liliaceae

♂ % P<sub>3+3</sub> A<sub>3+3</sub> G(3)



# Brassicaceae

♂ % K<sub>2+2</sub> Cx4 A<sub>2+4</sub>  
G(2)



Floral diagrams do not provide information about position of ovary.

Fruit

True

False

Parthenocarpic

Superior Ovary

Inferior Ovary

Fertilisation ✓

✓

x

Seeds ✓

✓

x

Eucarp

- Mango
- Coconut

Pseudocarp

- \* Pedicel - Chashewnut
- \* Thalamus - Apple
- \* Perianth - Mulberry (Morus alba)

Chemicals

- Auxin
- Gibberlin
- Banana (Natural)



→ Fruit (Ripened ovary)

→ Fruit wall / Pericarp

→ seed

- O Epicarp
- M Mesocarp
- I Endocarp

# Type of Fruit

Simple

Aggregate

Compound  
Multiple  
in fruits

Single flower

Single flower

Entire inflorescence

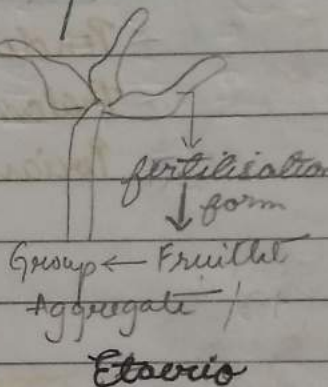
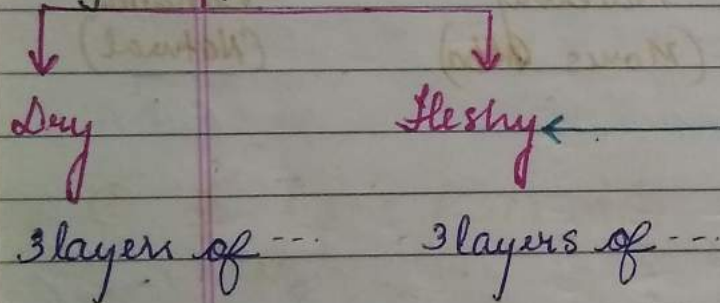
Gynoecium

Monotocarpellary

Apoecarpous

(1) Synchronous  
Hypanthodium  
Influence

Syncarpous



(2) Synchronous  
Spice  
Catkin  
Spadix

**DRY**

Pericarp and seed coat completely fused.

Caryopsis

Superior Gynoecium

Monocarpellary

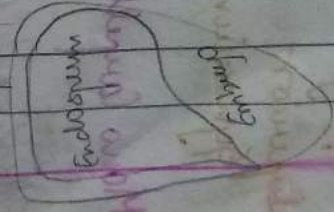
Poaceae family

Maize

Wheat

Rice

Pericarp



**Edible**

Entire fruit

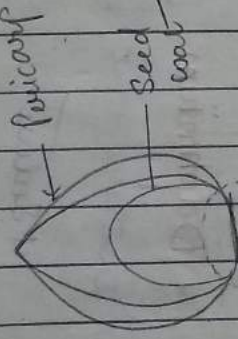
(ii) INDURSCICENT

Achene

Superior Gynoecium

Monocarpellary

Mirabilis



Fusion of Pericarp & Seed Coat

Restrictive to lower zone

Synsela

Superior Gynoecium

Bicarpellary

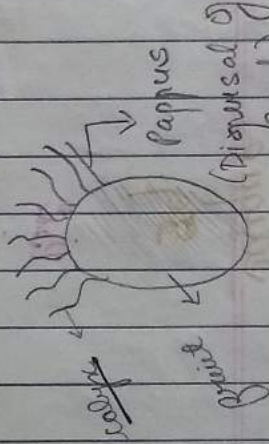
Asteraceae family

Sunflower

Sonchus

Lactuca

Zinnia



Paraschute Mechanism

Samaras

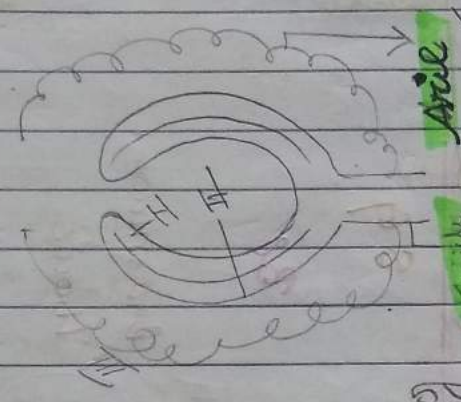
Superior Gynoecium

Winged fruit

from pericarp

Hiptage

Dioscorea



III Intermittent

OVULE

Nut (True)

Superior/Inferior Gynoecium

Mono-poly

**Pericarp**

Hard (stony)

woody

Cashewnut (Anacardium)

**Edible**

Thalamus & Cotyledons

Oak (Quercus)

Litchi (Marking nut)

**Aril / Edible**

3rd integument

arises from funicle

Simple: Fruit

# (ii) Dehiscent

Pod / legume

Superior  
Gynoceium

Mono Carpellary

Fabaceae / Pea  
Family

• Folicle

SG

Mono

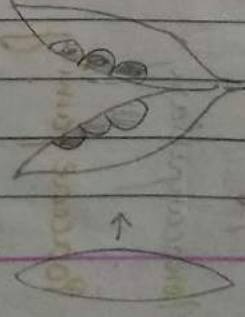
Dolichium

2 sutures + 1

1 Suture

unvalved  
in splitting

Etaria  
of  
Gallides



• 2 Sutures  
and both  
involved in  
splitting

Simple - dry

Siliqua

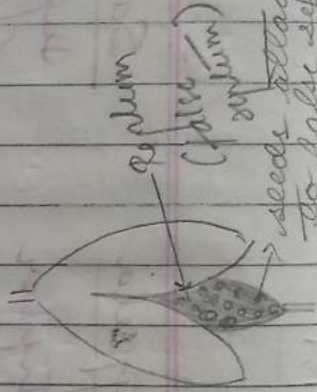
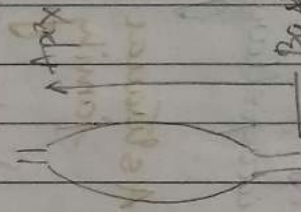
SG

Bi

Brassicaceae

Mustard

Radish



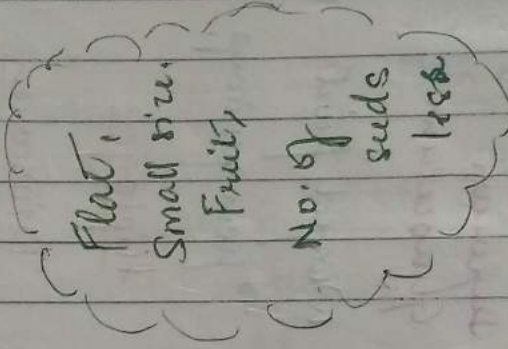
Silicle

SG

Bi

Brassicaceae

Therid



• Anemochory: seed dispersal due to air current

Capsule

SG

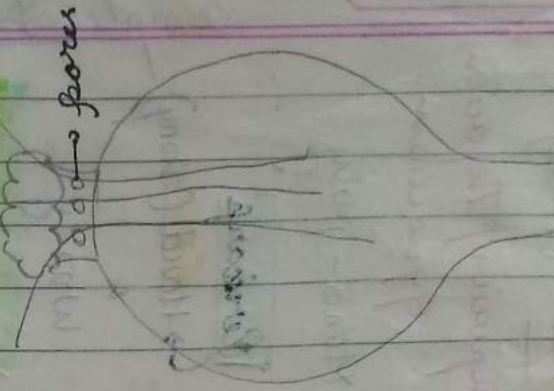
Bi - Poly

ovile glabrous

• Poppy

(Opium)

• → Datura



Anemochory  
Cinger Mem.

(1) SCITOCARPIC → Simple - Dry

Maturity

↓  
Fruit Breakage

↓  
Parts

Indehiscent Part is known

as Mericarp  
(Coriander)

Fruit

• Cremocarp

Infruct G.

Double Samara

SG.

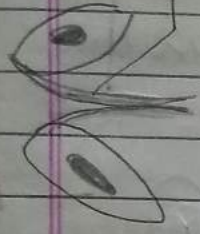
• Lomentum

Concavulus SG

Samara

$\frac{2}{4}$   
 $\times \frac{1}{4}$

→ Stylopodium  
(Remain of style)



Laripophore

coriander

→ Dehiscent parts known

as Cocci

Regma

Morei

3

Ricinus  
(Castor)

Example

Example

Cucurbitur

Acer

Tamarind

Fuls (Alumina)

Almond and walnut are fleshy fruit and not dry fruit.

Fleshy / Succulent

Drupe

• 3 layer of Pericarp

★ Endocarp - stony

Mango

Coconut

★ Walnut - Cotyledons  
Edible

Peach

Pear

Plum

Lizyphus

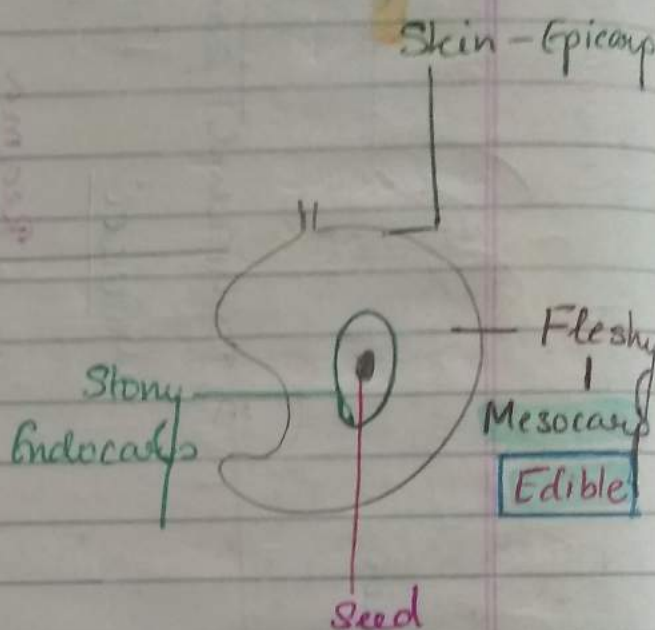
★ Almond

Epi - Velvety

Meso - Hard

Endo - Stony

Seeds - Edible



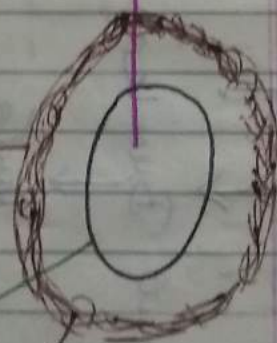
Edible - Endosperm  
Seed

Papery  
Epicarp

Stony  
Endocarp

Fibrous - Mesocarp  
Coke

★ Fibrous Drupe



## Berry

Superior Gynoecium

True Berry

Grapes

Tomato

Chillies

Beringal.

Inferior Gynoecium

False Berry

Guava

Banana

Edible - All parts are edible except Seeds

• Usually multiseeded

exception

- Beet nut  
Date palm

} Single  
Seeded.

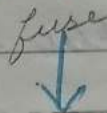
## • Pepe

(Hard Walled Berries)

Inferior Gynoecium

Cucurbitaceae

Epicarp + Thalamus



- Exocarp form the

- Skin of the fruit

and is Non edible

Cucumber - Edible - Mesocarp + Endo + Young Seeds

Bottle gourd - Edible - Mesocarp + Endo + Young Seeds

Watermelon - E - Endo + Placenta + ~~Meso~~  
~~non edible~~

Melon - E - Meso + Endo + ~~Placenta~~  
~~non edible~~

## • Pome

Inferior Gynoecium  
False / Pseudocarpic

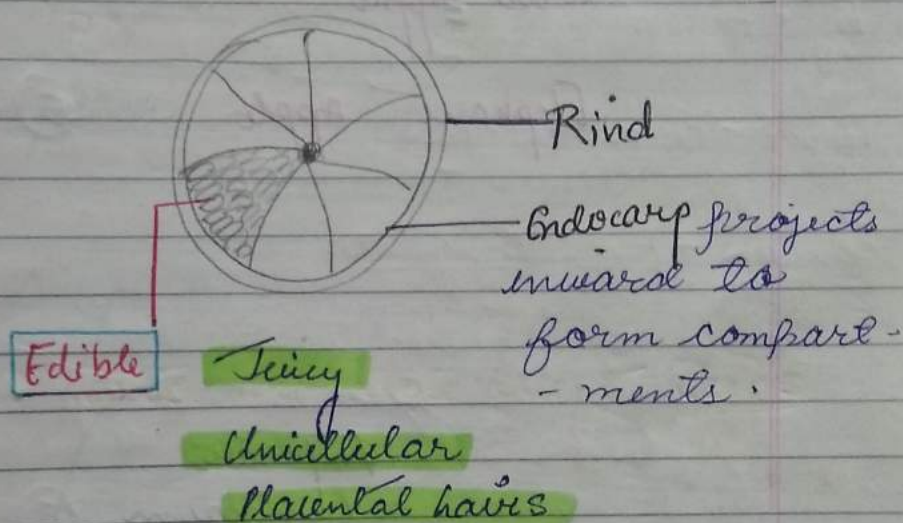
Apple } - Edible - Thalamus  
Pear }

• Hesperidium - Superior Gynoecium

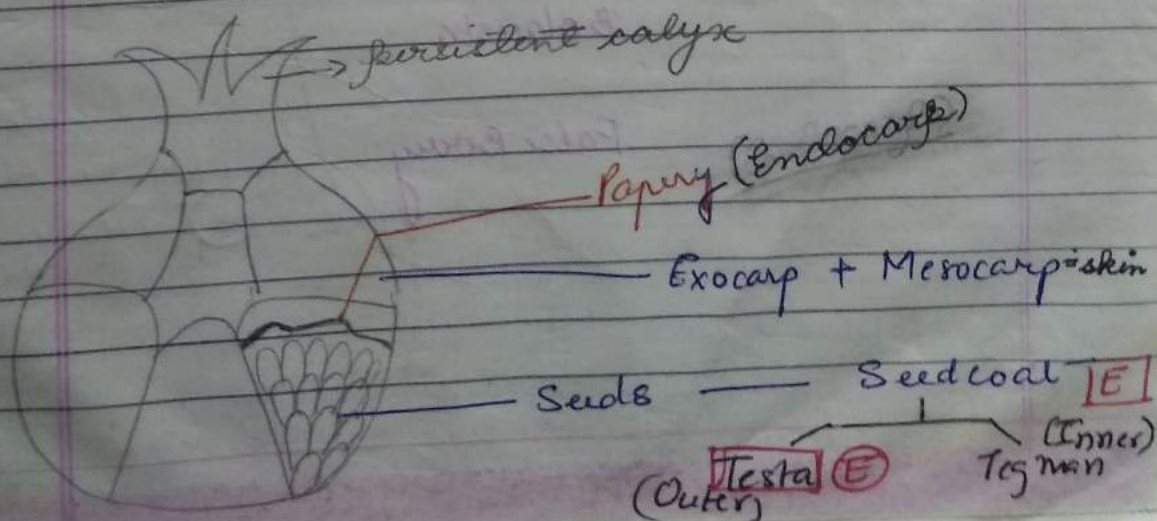
Epi + Mesocarp  
fused

↓  
Skin / Rind

Citrus



Balausta - Inferior Gynoecium  
Pomegranate



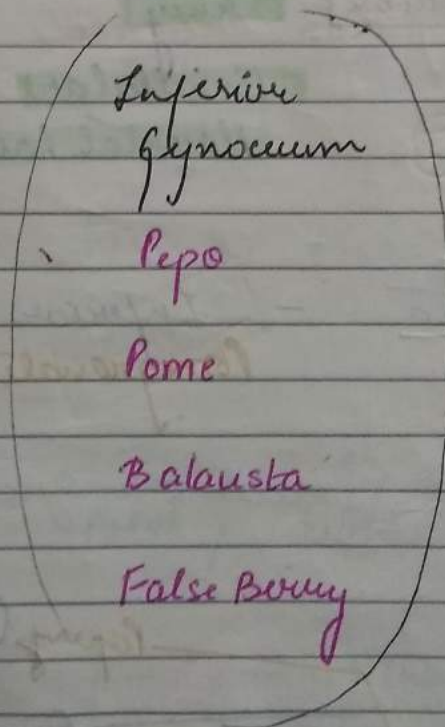
• Amphisarca - Superior Gynoecium

→ Epicarp - Stony

(E) - Mesocarp  
Endocarp  
Placenta

• Wood Apple

• Elephant Apple



Aggregat fruit

\* Elavio of Achenes - Strawberry  
(E) - thalamus

\* Elavio of berries - Custard Apple  
(E) inner layer of pericarp  
+ thalamus.

Elavio of follicles - Delphinium

\* Elavio of Drupe - Raspberry

\* after fertilisation female flower forms achene type of fruit.

## Composite / Multiple Fruit

↓  
Entire Inflorescence

### • Syconus

Hypanthodium

Fig - Ficus carica (E) - fleshy receptacle

Peepal - Ficus religiosa

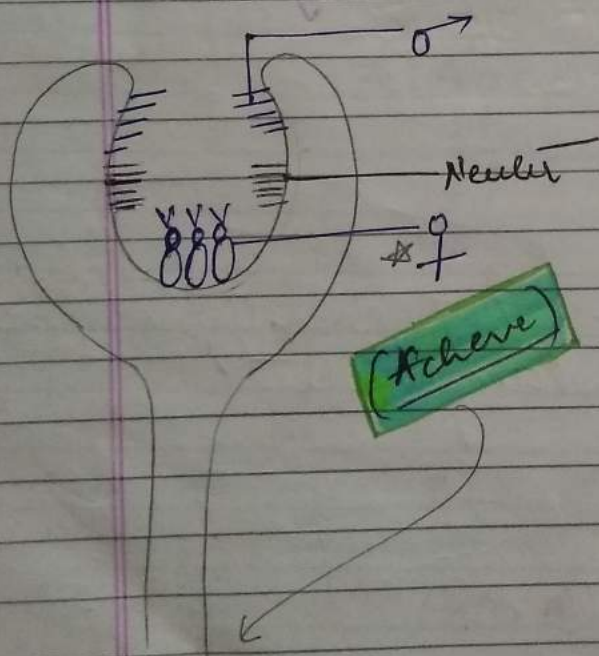
Banyan - Ficus bengalensis

### • Sorbus

Sipke, Caltin,  
Spadix

Pineapple

(E) - Fleshy axis +  
Bract +  
Perianth +  
pericarp



### Jackfruit

(E) Fleshy axis +  
Bract +  
Perianth +  
pericarp +  
Seeds

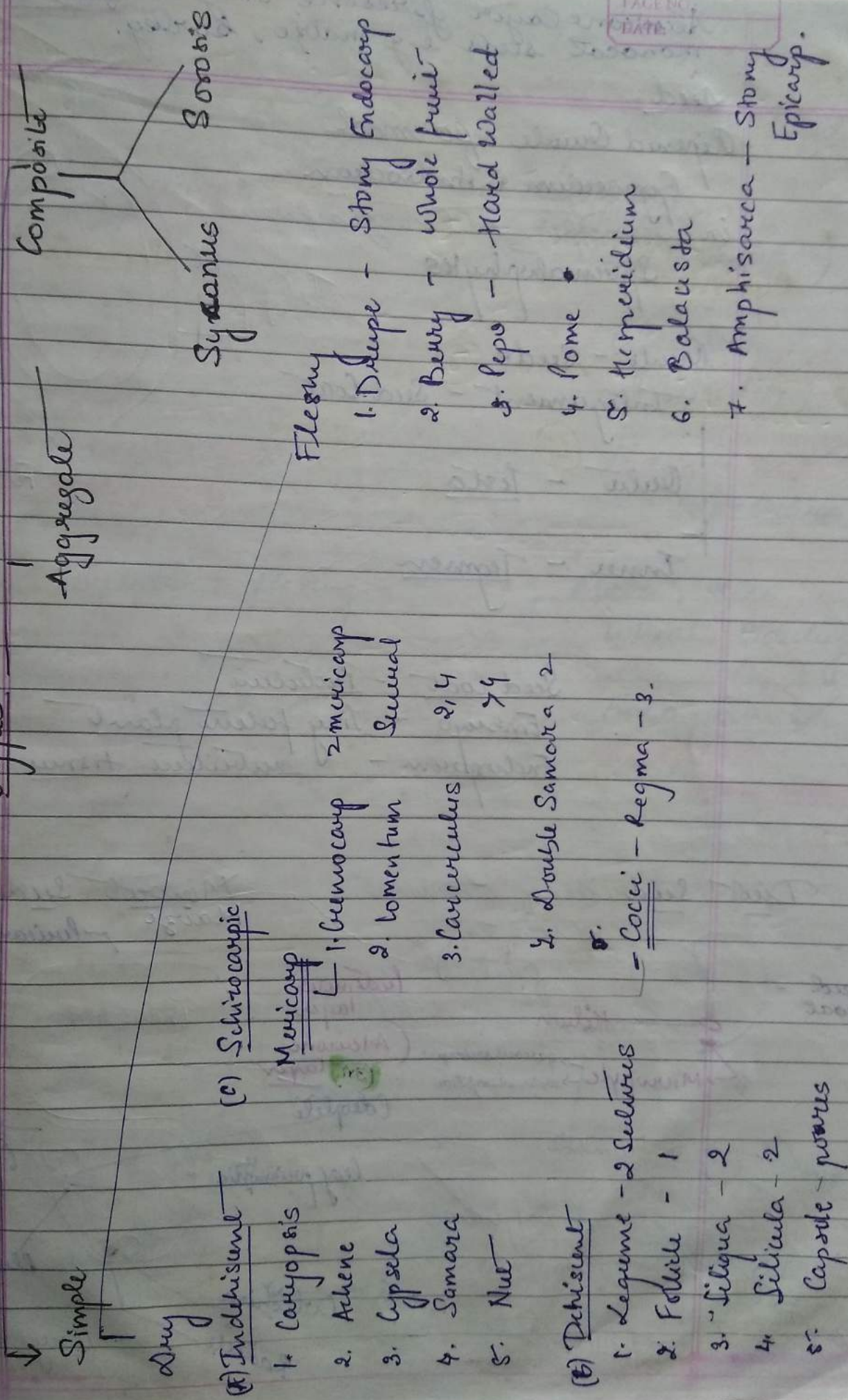
### Mulberry

(Morus Alba)

(E) Fleshy axis +  
Perianth

In syconous fruit  
the female flowers  
form achene type  
of fruit.

# Types



Aleurone layer present only in few monocot seeds e.g. maize, barley.

## Seed

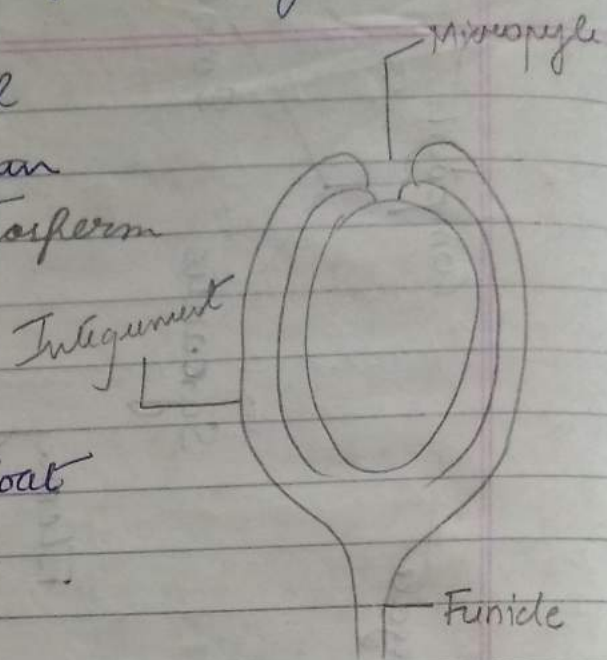
Ripened Ovule formed  
Gynoecium + Androecium  
in Gymnosperm and Angiosperm  
Spermatophytes

Ovule - Seeds

Integument - Seed Coat

Ovule - Testa

Inner - Tegmen

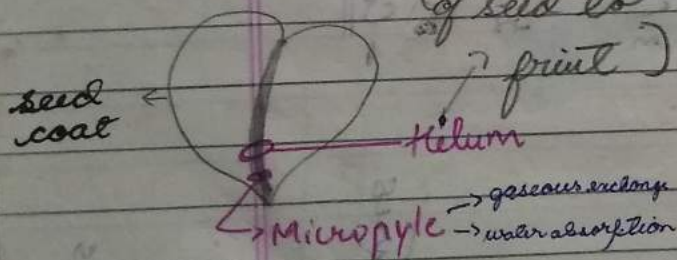


Seed Coat - Protective

Embryo - tiny future plant

Endosperm - nutritive tissue

Dist Seed (pt. of attachment  
of seed to  
fruit)



Monocot Seed -  
Maize - Pericarp +

Seed

Coat

Protective layer  
(Aleurone  
3n layer)  
Coleoptile

Endosperm

Scutellum  
(Shield  
Shape)

Leaf primordia

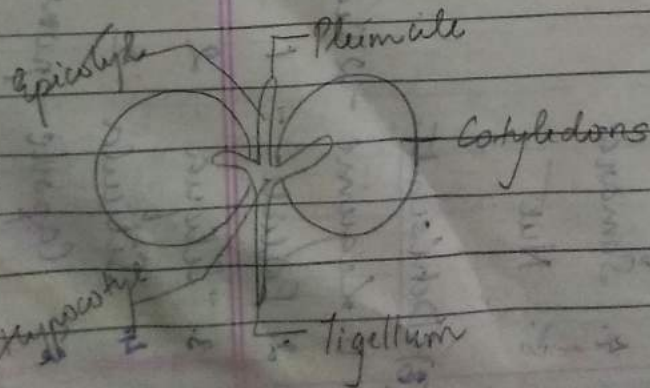
Plumule

Coleoptile

Radicle

Shield  
Shape

Caryopsis fruit



Castor has : - Endosperm  
- Perisperm  
- Caruncle.

Cyanobium

Dicot Seed

Monocot Seed.

2 Cotyledons

Scutellum  
(1 cotyledon)

X

Coleoptile  
Cotyledon

Ex albuminous

Endosperm X

Exception: Castor

Albuminous

Endosperm ✓ seeds

Exp - Orchids

Cotyledons fleshy

e.g Maize, Rice,  
Wheat, Barley

e.g Grams, Pea, Bean  
Groundnut

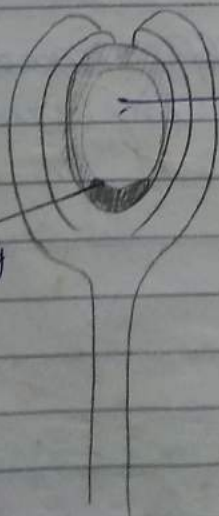
Endosperm X

Exception :

Orchids

Perisperm

Modification  
of style.



Nucellus

Seed (-)

Seed (+)  
remains

Perisperm

(1) Gymnosperm

Pinus

(2) Angiosperm

Piper nigrum  
Castor

Chalazium  
nucellus present only  
at base

e.g → Gynostemium

- Tendril in Antigonon is modification of inflorescence.
- Cytoplasmic male sterility gene affects staminode, sterile stamen which is characteristic feature of caesalpinaceae.
- Important chemicals and families
  - Singrin — Brassicaceae
  - Carthamin — Asteraceae
  - Atropine — Solanaceae
  - Alcin — Liliaceae
- Primitive family (dicot) — Ranunculaceae
- Advanced family (dicot) — Asteraceae.
- Anthesis  $\rightarrow$  Opening of flower bud.
- Geocarpic fruit  $\rightarrow$  ground nut
- Single flower appearance in cyathium and capitulum inflorescence.